

**SUMMER INTERNSHIP REPORT ON
FUNDAMENTAL ANALYSIS OF IT SECTOR COMPANIES**

**Submitted in partial fulfilment for the award of the degree of
MASTER OF MANAGEMENT STUDIES
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**Submitted by
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**Under The Guidance of
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CERTIFICATE

This is to certify that the summer internship report titled “Fundamental Analysis of Public Sector Banking” submitted in partial fulfilment for MMS Degree Examination by **Rashmi Gupta** is a record of Research work carried out by the student during the period from **2023 to 2024** under my guidance, and has been found satisfactory. This project work is original and not submitted earlier for the award of any degree, diploma or associateship of any other University/Institute.

COLLEGE STAMP HERE

Dr. Rasika Mallya

DIRECTOR

Dr. Rahul N. Wadekar

PROJECT GUIDE

EXTERNAL EXAMINER



EXPERIENCE / RELIEVING CERTIFICATE

This is to certify that **Ms. Rashmi Gupta** has completed her internship as "**Finance Accounting Executive**" with **J C AGAL & COMPANY** from 1st May, 2024 to 31st July, 2024.

As an intern of our company, Her performance is satisfactory to the work she had been assigned, Her dedication and determination have proven.

She has successfully completed her summer internship with us . We wish him all the best for his future endeavors.

With Regards,

From

CA Jagdish C Agal

Proprietor



J C Agal & Company Mumbai.

CA Jagdish Agal

Date 31.07.2024

DECLARATION

I hereby declare that this project report submitted by me to the partial fulfillment of the requirement for the award of **Master of Management Studies** (MMS) of the University of Mumbai is a bonafide work undertaken by me and it is not submitted to any other University or Institute for the award of any degree, diploma/ certificate or published any time before.

Name : Rashmi Gupta

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Signature

(Rashmi Gupta)

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EXECUTIVE SUMMARY

Summer training is crucial for management students, transforming them into well-rounded professionals.

It bridges the gap between theoretical knowledge and practical application in real-world scenarios.

This project, titled “Fundamental Analysis of the Private Sector in Information Technology,” focuses on determining the intrinsic value of securities by evaluating various economic and financial factors. This analysis is particularly valuable for investors seeking opportunities in private IT companies.

In today's competitive business environment, the IT sector in India stands out for its intense competition.

Key financial statements used in this sector include balance sheets, which depict a company's financial position, and profit & loss statements, which reveal net profits and losses.

Ratio analysis is a widely used method for evaluating an IT company's performance over time. In my report, I analyzed the financial statements of private sector IT Companies over the past five years.

Through this process, I learned that ratio analysis is instrumental in identifying past shortcomings and informing future strategies.

Hence, it is essential for every organization, regardless of size, to prepare financial statements and analyze them using ratios. This approach not only enhances understanding of the company's financial health but also aids in making informed decisions for the future.

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CHAPTER 1

INTRODUCTION

1.1 NEED OF THE STUDY

- To gain exposure to the corporate world.
- To understand the IT sector fundamentally.
- To know how economic changes, natural calamities, and crises impact the industry.
- To predict the future of the IT sector with the help of research, fundamental, and economic analysis.
- To study the ratio analysis of selected IT companies.

1.2 OBJECTIVE OF THE STUDY

- To study the changes the organization has undergone in the recent past or have initiated recently.
- To study fundamental and technical analysis.
- To study the impact of post-COVID on the private IT sector.

1.3 SCOPE OF THE STUDY

This project will allow investors to refer to the performance of IT companies, their relative growth, and decide whether to invest in particular shares. The project is limited to the fundamental analysis of the selected IT companies for a given period; other analyses and IT companies not present in this list are not considered.

CHAPTER 2

ORGANIZATIONAL BACKGROUND



2.1 ORGANISATIONAL INTRODUCTION

M/s J.C. AGAL & CO. is a Chartered Accountant firm established in 1966, recognized as Category-II by RBI, and based in Mumbai. J.C. AGAL & CO. offers consultancy solutions in Direct Tax, Indirect Tax, Company Law Matters, Auditing, Accounting, and Financial Services. They also provide services related to Company Registration, Joint Ventures, LLP Registration, VAT/CST Registration, Service Tax Registration, Import Export Code, International Taxation, Management Consultancy, and Due Diligence Services.

J.C. AGAL & CO. emphasizes understanding clients' needs to offer tailored financial solutions. The firm employs experienced and qualified professionals, ensuring integrity and independence in their services. Continuous professional development is a key focus, achieved through meetings, group discussions, and participation in seminars.

The firm's head office in South Mumbai is well-equipped with modern technology. The team consists of over thirty staff members, including M.Com and B.Com graduates and articles.

Scope of Services Offered:

- Professional Services: Audit, Taxation, Company set-up and registration, financial recruitment, and legal services including patent and trademark registration.
- Financial Accounting: Outsourcing services such as bookkeeping, payroll, office administration, and CFO services.
- Management Accounting: Services in budgeting, forecasting, cash flow management, and business strategy.
- Systems Solutions: Efficiency improvement services including management information systems implementation.
- Business Management: Consulting and corporate finance services including M&A, business due diligence, and market entry strategy.

Professional Services:

- Audit: Comprehensive audit services for local and international reporting, including GAAP and IAS compliance.
- Taxation: Extensive tax services for both foreign and domestic enterprises, including compliance and tax optimization.
- Company Registration: Assistance in setting up various types of business entities and ensuring compliance with Indian regulations.
- Legal Services: Through an affiliate legal team, PPKC provides high-quality legal services essential for businesses in India.

Financial Accounting:

- CFO Services: Providing experienced personnel to assist with financial strategy and management.
- Bookkeeping: Outsource bookkeeping services to ensure compliance and add value through quality management accounting information.
- Accounting and Financial Advice: Expert knowledge on accounting and financial issues, with experience in both international and Indian GAAP.

Management Accounting:

- Budgeting and Forecasting: Comprehensive financial modeling to guide businesses in achieving financial goals.
 - Cash Flow Management: Proven systems to help manage finances, including account receivables and payables.
- Market Growth Strategy: Assisting firms in setting effective strategies for growth in the Indian market.

Systems Solutions:

- Systems Implementation: Choosing and implementing the most appropriate financial, HR, and accounting solutions.
- Information Management: Streamlining information flows to enhance cost competitiveness.
- Process Re-engineering: Ensuring informed business decision-making through thorough understanding of business drivers.

Business Management:

- Business Due Diligence: Comprehensive evaluation of acquisition targets and existing operations.
- M&A and Divestitures: Guiding projects from conception to execution, including synergy optimization and business development.
- M&A Integration: Top-down assessment of integration processes to achieve optimal synergies.

Clients:

J.C. AGAL & CO. has handled taxation matters, internal control and system checks, statutory audits, concurrent audits, revenue audits, stock audits, and portfolio audits for a wide range of clients, including prominent names like Wipro Yardley Consumer Care Pvt. Ltd., The Hotel LaLit, and various public sector banks.

This profile captures the essence of J.C. AGAL & CO. ,emphasizing their expertise, services, and client engagements in a concise manner.

2.2 SERVICES

They are leaders Taxation laws, Accounting Advisory and Financial services.

The major activities of this company as per the listed official records are Accounting and Book-keeping activities, auditing, tax consultancy, services.

2.3 MISSION AND VISION

- ❖ Mission - We believe that our experience balances the requirements of the clients and our team will put their best selves into solving tough problems that require knowledge and insight. A team of committed professionals, we provide tailor made financial solutions for each of our clients. With our dedication to personalized service and adding real value, we work closely with you to understand your individual business and unique business requirements, allowing our experts to offer you the most up-to-date and expert advice for all of your needs
- ❖ Vision - We believe that the strength of professional service depends upon the quality of professional staff. Keeping in view this, we employ a team of well-experienced and qualified people who have attained that level of mental maturity, which ensures professional integrity and independence. Our senior level staffs have worked on audit, management and financial assignments in different fields, and are fully conversant with accounting principles and methods used in these fields.

2.4 DEPARTMENTS

1. Professional Services: Audit, Taxation, Company set-up and registration, professional financial recruitment and legal services covering patent and trademark registration, business contracts and employment contracts.
2. Financial Accounting: outsource accounting services from the provision of CFO services to bookkeeping, payroll and office administration, including company policy and procedure manuals.
3. Management Accounting: market growth related services including training and assistance in budgeting, forecasting, cash flow management and business strategy.

4. Systems Solutions: efficiency improvement services, including review and implementation of management information systems and surrounding processes.
5. Business Management: management consulting and corporate finance services, including assistance with market entry strategy, mergers, acquisitions and divestitures, business and financial due diligence, business investigation, internal communication and M&A integration.

JOB DISCRIPTION

- Prepared financial statements, including Income Statements, Balance Sheets, and Cash Flow Statements.
- Prepared contracts and documents for transactions.
- Managed and updated the client database and contact information.
- Prepared computations for income tax under various heads
- Scheduled and coordinated client meetings

CHAPTER 3

FUNDAMENTAL ANALYSIS

1.1 INTRODUCTION TO THE EQUITY

- What is equity share ?

An equity share, normally known as ordinary share is a part ownership where each member is a fractional owner and initiates the maximum entrepreneurial liability related to a trading concern. Equity shares are long-term financing sources for any company. These shares are issued to the general public and are non-redeemable in nature. Investors in such shares hold the right to vote, share profits and claim assets of a company.

- What is Equity Research ?

Equity research is the study of a business and its environment to make a buy or sell decision about investing in its shares. Equity research is all about producing research reports, price projections, and recommendations of stocks based on proper research and study. Equity research professionals are responsible for producing analyses, recommendations, and reports on investment opportunities that investment banks, institutions, or their clients may be interested in.

- How do prices of equity shares move ?

Stock prices are determined in the marketplace, where seller supply meets buyer demand. Stock prices are driven by a variety of factors, but ultimately the price at any given moment is due to the supply and demand at that point in time in the market. Fundamental factors drive stock prices based on a company's earnings and profitability from producing and selling goods and services. Technical factors relate to a stock's price history in the market pertaining to chart patterns, momentum, and behavioural factors of traders and investors.

1.2 IT SECTOR

IT sector stands out for its vibrant and rapidly evolving nature. This industry includes companies involved in technology and software development, IT services, and consulting. Unlike public sector entities, which are managed by the government, IT firms are usually privately owned or publicly traded, focusing on maximizing profitability and fostering innovation. These companies are at the forefront of technological advancements, often investing heavily in research and development to stay competitive. When analyzing IT companies, it's essential to review their financial performance, including metrics like revenue growth, profit margins, and return on investment. Additionally, understanding the impact of market trends and economic conditions on the IT sector can provide valuable insights into a company's potential for growth and sustainability. This comprehensive analysis helps investors assess the intrinsic value of IT companies and make informed investment choices. These companies span a broad spectrum, from software development and IT services to consulting. To truly understand their value, it's crucial to look at their financial health, including revenue growth and profit margins, while also considering their commitment to innovation through R&D investments. Additionally, keeping an eye on market trends and economic factors helps paint a complete picture of how these companies are positioned in a competitive landscape. This thorough approach helps investors make well-informed decisions about the true worth of IT companies.

ADVANTAGES

- High growth potential due to increasing reliance on technology.
- Innovation-driven with continuous advancements and new product developments.
- Diverse revenue streams, including software sales, consulting services, and cloud computing.
- Global market reach, allowing access to a vast customer base.
- Scalability of solutions, particularly in software and cloud services.
- High profit margins, especially in software and cloud services.
- Job creation and economic growth through high-skilled employment.
- Strong consumer demand for the latest technology products and services.

DISADVANTAGES

- Rapid obsolescence of products and services.
- Intense competition squeezing profit margins.
- High costs of research and development.
- Significant cybersecurity risks and data privacy concerns.
- Complex and evolving regulatory challenges.
- Market volatility affecting stock prices and financial performance.
- Talent shortages impacting growth and innovation.
- Dependency on economic conditions, which can affect IT spending.

❖ Nationalization

The IT industry in India, which includes tech services and business outsourcing, plays a big role in the country's economy. In the fiscal year 2022, this sector contributed 7.4% to India's GDP. For the fiscal year 2023, the industry's total revenue was around \$245 billion. Out of this, \$51 billion came from within India, and \$194 billion came from exports.

The industry provides jobs to about 5.4 million people as of March 2023. In December 2022, it was reported that Indian IT companies exported software worth around Rs 11.59 lakh crore in the previous year (2021-22).

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❖ Expansion

India's IT industry has been growing a lot lately:

1. Making More Money: The industry is raking in a lot of cash—about \$245 billion in 2023.
2. Creating Jobs: It's a big job creator, with around 5.4 million people working in it.
3. Exporting a Lot: India sells a huge amount of IT services to other countries, about \$194 billion worth.
4. Strong Local Market: It also earns around \$51 billion from services within India.
5. Cutting-Edge Tech: The sector is always on the move, bringing in new technologies like AI and cloud computing.
6. Growing Infrastructure: Investments in tech hubs and special zones are helping companies expand.
7. Government Support: The government provides support and incentives, making it easier for the industry to grow.

In simple terms, India's IT industry is booming, creating jobs, making a lot of money, and becoming a big player globally.

❖ Priority Sector Lending

Priority Sector Lending (PSL) in India includes some support for the IT sector in these ways:

Loans for IT Startups: Small tech companies and startups can get loans to help them grow and develop new technologies.

Tech for Financial Access: PSL supports technology that helps bring banking and financial services to more people, especially in remote or underserved areas.

Educational Tech: Funding can be available for tech tools that improve education, like online learning platforms.

Infrastructure Support: Investments in tech-related infrastructure, like tech parks or innovation centers, may get some indirect support.

❖ Social Development Initiatives

Online Learning: Platforms like Coursera make education accessible to everyone, including those in remote areas.

Telemedicine: Apps like Practo enable remote medical consultations, enhancing healthcare access.

E-Governance: Digital services simplify interactions with government agencies, such as applying for permits online.

Mobile Banking: Apps like Google Pay provide financial services to underserved areas.

Smart Cities: Tech improves urban living through efficient traffic management and energy use.

Digital Skills Training: Programs like Google's Digital Garage boost digital literacy and career prospects.

Overall, IT improves education, healthcare, governance, and financial inclusion.

❖ Reforms and consolidation

Reforms and consolidation in the IT sector focus on improving efficiency and adapting to new challenges. Key reforms include updating regulations for data protection and tech innovation, such as GDPR. Infrastructure upgrades involve investing in cloud computing and faster networks. Mergers and acquisitions help companies expand and streamline operations. Standardizing processes enhances system interoperability. Enhanced cybersecurity measures protect against emerging threats. Additionally, initiatives to develop digital skills ensure the workforce is prepared for new technologies. These efforts aim to make the IT sector more robust, secure, and innovative, driving growth and adaptability in a rapidly evolving landscape .

❖ Technological Advancements

Technological advancements in IT are revolutionizing industries. **Artificial Intelligence (AI)** drives automation and data insights. **Cloud Computing** provides scalable storage and computing power. **Big Data Analytics** extracts valuable insights from large datasets. **5G Technology** offers faster speeds and lower latency, enhancing IoT and real-time applications. **Cybersecurity** innovations protect against evolving threats. **Blockchain** ensures secure, transparent transactions. **Internet of Things (IoT)** connects devices for smarter homes and industries. **Quantum Computing**, though emerging, promises breakthroughs in complex problem-solving. These advancements enhance efficiency, drive innovation, and open new opportunities across various sectors, transforming how we live and work.

❖ Government Ownership

Government ownership in IT involves direct control and influence over technology and services. Governments may own or have stakes in public IT companies, manage critical infrastructure like national data centers and cloud services, and set regulations to ensure data protection and fair practices. They also develop and own digital platforms for public services, such as e-governance and online tax systems. Additionally, governments fund and support IT research and innovation to drive technological advancement and digital inclusion. This ownership helps ensure the security, availability, and efficiency of essential digital services while aligning with national goals and policies.

1.3 IMPORTANCE OF IT SECTORS

1. Efficiency and Productivity:

- **Automation:** IT enables automation of repetitive tasks, streamlining processes and improving efficiency.
- **Data-Driven Decision Making:** IT systems collect, analyze, and interpret data, providing valuable insights for informed decision-making.
- **Supply Chain Management:** IT solutions optimize supply chain management, ensuring efficient flow of goods and services.

2. Communication and Collaboration:

- **Internal and External Communication:** IT facilitates seamless communication and collaboration within organizations and with external stakeholders.
- **Remote Work:** IT enables remote work, allowing for increased flexibility and productivity.
- **Global Connectivity:** IT connects businesses and individuals worldwide, fostering global collaboration and trade.

3. Innovation and Product Development:

- **Research and Development:** IT tools and platforms support research and development, enabling the creation of new products and services.
- **Product Design and Development:** IT software aids in product design, prototyping, and testing.
- **Customer Experience:** IT enables personalized customer experiences through targeted marketing and customer relationship management.

4. Cybersecurity and Risk Management:

- **Data Protection:** IT systems protect sensitive data from unauthorized access and cyber threats.
- **Risk Assessment and Mitigation:** IT tools help identify and mitigate risks, ensuring business continuity and compliance with regulations.

5. Financial Management:

- **Accounting and Financial Reporting:** IT systems automate accounting processes and generate accurate financial reports.
- **Payment Processing:** IT enables secure and efficient payment processing, both online and offline.
- **Financial Analysis:** IT tools analyze financial data to identify trends, assess performance, and make informed financial decisions.

6. Customer Relationship Management (CRM):

- **Customer Data Management:** IT systems store and manage customer data, enabling personalized interactions and targeted marketing.
- **Sales and Marketing Automation:** IT automates sales and marketing processes, improving

efficiency and customer satisfaction.

- **Customer Service:** IT tools support customer service operations, providing efficient and effective assistance.

7. Human Resources Management:

- **Employee Data Management:** IT systems store and manage employee data, including payroll, benefits, and performance information.
- **Recruitment and Onboarding:** IT tools streamline recruitment and onboarding processes, improving efficiency and reducing costs.
- **Performance Management:** IT systems support performance management, including goal setting, performance reviews, and talent development.

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1.4 12 TOP COMPANIES OF IT SECTORS

IT companies can be owned by individuals, private companies, public companies, government agencies, or non-profits. Listed below are all the Indian IT sector with their headquarters, year of establishment.

Sr No.	Name	Headquarter	Establishment Year	Government shareholding
1.	Tata Consultancy Services	Mumbai. Maharashtra	1968	57.56%
2.	Infosys	Pune, Maharashtra	1981	81.40%
3.	Capgemini	Pune, Maharashtra	2000	90.97%
4.	Cognizant Technology Solution	Chennai, Tamil Nadu	1994	93.08%
5.	Wipro	Bengaluru, Karnataka	1945	62.93%
6.	Tech Mahindra	Pune, Maharashtra	1986	79.86%
7.	Oracle	Bengaluru, Karnataka	1997	96.38%
8.	HCLTech	Noida. India	1976	73.15%

9.	Deloitte	Hyderabad	1990	97.07%
10.	KPMG	Mumbai, Maharashtra	1908	63.97%
11.	Accenture	Mumbai, Maharashtra	1919	83.49%
12.	Tata Communication	Mumbai, Maharashtra	1943	95.39%

1.5 CHALLENGES OF IT SECTOR COMPANIES

- **1. Rapid Technological Advancements:**
person running on a treadmill with a rapidly changing background
- **Challenge:** Keeping up with the constant evolution of technology can make it difficult to accurately assess the long-term value of IT companies.
- **Impact:** Outdated analysis can lead to inaccurate valuations and missed investment opportunities.
- **2. Intellectual Property:**
- light bulb with a dollar sign inside
- **Challenge:** Valuing intangible assets like patents, copyrights, and trademarks can be complex and subjective.
- **Impact:** Overreliance on tangible assets can undervalue IT companies with significant intellectual property.
- **3. Future Earnings Projection**
- person looking into a crystal ball
- **Challenge:** Predicting future earnings growth in the IT sector can be challenging due to rapid technological changes and intense competition.
- **Impact:** Inaccurate earnings projections can lead to overvalued or undervalued stocks.
- **4. Competitive Landscape:**

- **Challenge:** The IT sector is highly competitive, with new entrants and disruptive technologies emerging constantly.
- **Impact:** Assessing a company's competitive advantage and market position can be difficult.
- **5. Valuation Metrics:**
 - person looking at a financial chart with various metrics
 - **Challenge:** Traditional valuation metrics like P/E ratios may not be as effective in the IT sector due to intangible assets and rapid growth.
 - **Impact:** Investors may need to explore alternative valuation methods, such as discounted cash flow analysis or comparable company analysis.
- **6. Market Sentiment and Speculation:**
 - person looking at a stock market chart with fluctuating lines
 - **Challenge:** Market sentiment and speculation can significantly influence IT stock prices, sometimes leading to irrational valuations.
 - **Impact:** Fundamental analysis may not always accurately predict short-term price movements.
- **7. Regulatory Changes:**
 - person reading a legal document
 - **Challenge:** Changes in regulations, such as tax laws or privacy regulations, can have a significant impact on IT companies.
 - **Impact:** Investors must stay updated on regulatory changes and their potential implications for the industry.

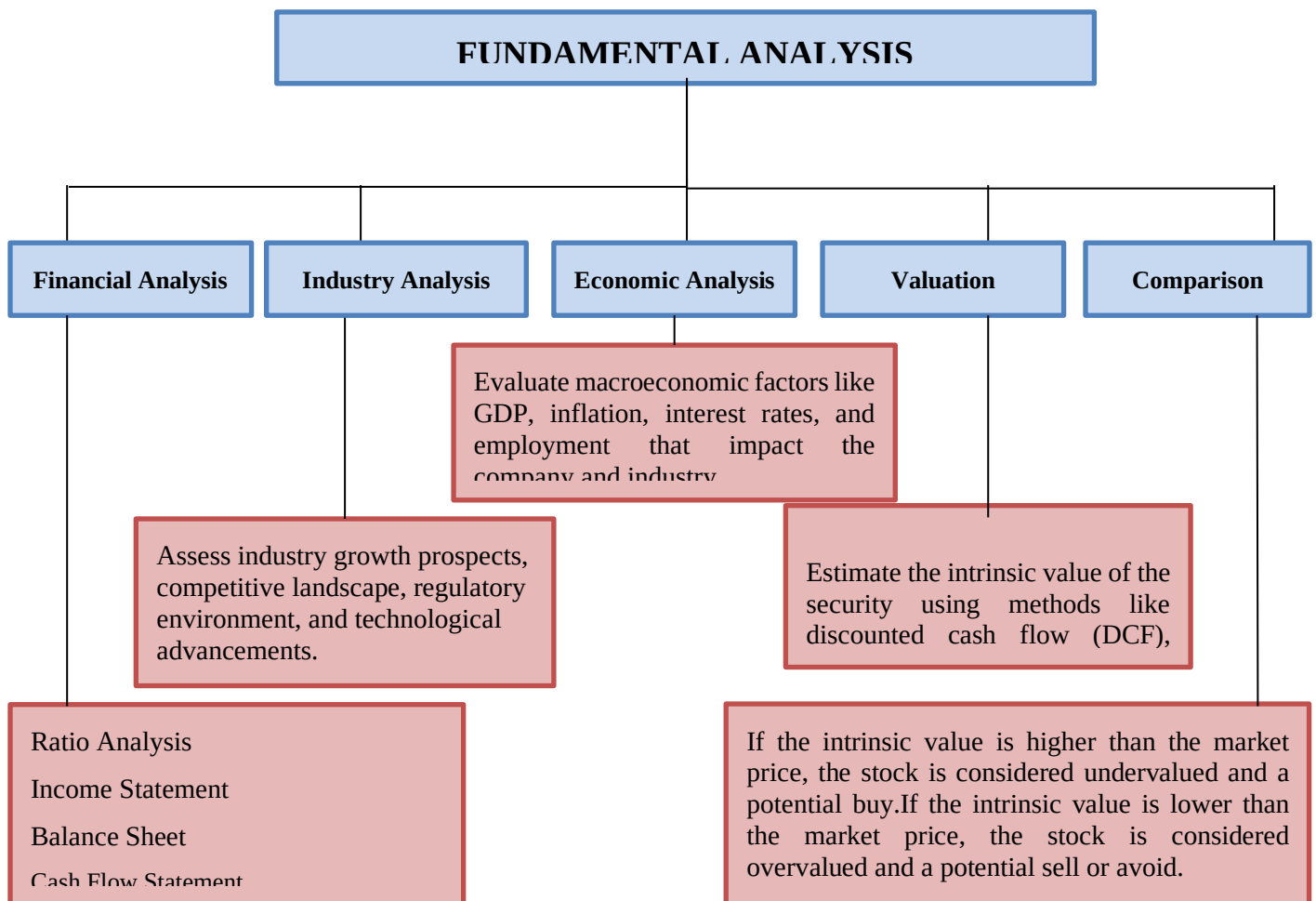
1.6 IMPACT OF COVID 19 IN IT SECTORS

- The COVID-19 pandemic significantly impacted the financial analysis of IT companies, requiring analysts to adapt their methodologies and consider new factors. Here are some key areas affected:
 - Revenue and Profitability:
- Increased Revenue from Digital Services: Many IT companies experienced increased revenue from digital services, such as cloud computing, cybersecurity, and e-commerce solutions, as businesses accelerated their digital transformation.
- Decreased Revenue from Traditional Services: Some traditional IT services, like on-premise software and hardware, faced a decline in demand due to economic uncertainty and remote work trends.
- Profit Margin Analysis: Analysts had to carefully evaluate profit margins to assess the impact of increased costs (e.g., remote work expenses, cybersecurity investments) on profitability.
 - Cash Flow Analysis:
- Working Capital Management: The pandemic highlighted the importance of efficient working capital management, as businesses faced disruptions in supply chains and customer payments.
- Cash Burn Rate: Analysts closely monitored cash burn rates to assess the financial health of companies, especially those with high growth rates.
 - Risk Assessment:
- Cybersecurity Risks: The increased reliance on digital technologies highlighted the importance of assessing cybersecurity risks and their potential impact on financial performance.
- Supply Chain Risks: Disruptions in global supply chains posed significant risks to IT companies, requiring careful analysis of their exposure and mitigation strategies.
 - Valuation Metrics:
- Traditional Valuation Metrics: Traditional valuation metrics like P/E ratios and price-to-book ratios may not be as effective in the post-pandemic era, as the pandemic disrupted traditional business models and growth prospects.
- Alternative Valuation Methods: Analysts had to explore alternative valuation methods, such as discounted cash flow analysis and comparable company analysis, to account for the unique challenges and opportunities presented by the pandemic.
 - Future Outlook:
- Long-Term Growth Prospects: Analysts had to reassess the long-term growth prospects of IT companies, considering the potential impact of the pandemic on industry trends, competitive landscapes, and regulatory environments.
- Overall, the COVID-19 pandemic forced analysts to adapt their financial analysis methodologies to account for the unique challenges and opportunities presented by the crisis. By carefully considering the factors mentioned above, analysts can make more informed assessments of the financial health and future prospects of IT companies.

1.7 FUNDAMENTAL ANALYSIS

The various factors of interest in fundamental analysis can be broadly classified into three categories – economy wide factors, Industry wide factors and Company wide factors. Hence, we have,

1. **Economic Analysis**
2. **Industry Analysis**
3. **Economy Analysis**
4. **Valuation Analysis**
5. **Comparison Analysis**



Advantages of fundamental analysis :

- **Intrinsic Value Assessment:** Helps determine a security's true worth based on underlying assets and future earnings.
- **Long-Term Focus:** Emphasizes the company's long-term health and growth potential.
- **Risk Evaluation:** Assesses risks associated with an investment, such as financial stability and industry outlook.
- **Dividend Analysis:** Evaluates the sustainability and growth potential of dividends for dividend-paying stocks.
- **Company-Specific Insights:** Provides a deep understanding of a company's business, operations, and management.
- **Active Investment Strategy:** Supports active investors seeking undervalued or overvalued securities.

Limitations fundamental analysis

- **Time-Consuming**
Requires extensive research and analysis.
- **Requires Expertise**
Demands a high level of knowledge and understanding.
- **Market Sentiment Impact**
May be influenced by market sentiment and external factors.

Disadvantages of fundamental analysis :

- **Time-Consuming**
Requires extensive research and analysis.
- **Requires Expertise**
Demands a high level of financial knowledge.
- **Subjectivity**
Involves subjective judgments, such as management assessment.
- **Market Sentiment Impact**
May be influenced by short-term market fluctuations.
- **Economic Uncertainty**
Can be affected by rapidly changing economic conditions.

- **Information Asymmetry**

May be limited by unequal access to information.

Tools of Fundamental Analysis

Fundamental analysis involves evaluating a company's financial health and prospects to determine its intrinsic value. Here are some of the key tools used in fundamental analysis:

Financial Statements

- **Income Statement**

Shows a company's revenues, expenses, and net income over a specific period.

- **Balance Sheet**

Provides a snapshot of a company's assets, liabilities, and equity at a particular point in time.

- **Cash Flow Statement**

Reveals a company's cash inflows and outflows from operating, investing, and financing activities.

Financial Ratios

- **Profitability Ratios**

Measure a company's ability to generate profits. Examples include profit margin, return on assets, and return on equity.

- **Liquidity Ratios**

Assess a company's ability to meet its short-term obligations. Examples include current ratio and quick ratio.

- **Solvency Ratios:** Evaluate a company's long-term financial stability. Examples include debt-to-equity ratio and interest coverage ratio.

- **Efficiency Ratios**

Measure how efficiently a company uses its assets and resources. Examples include inventory turnover ratio and asset turnover ratio.

Industry Analysis

- **Industry Trends**

Understanding industry trends can help assess a company's growth potential and competitive landscape.

- **Competitive Analysis**

Comparing a company to its peers can reveal its strengths and weaknesses.

Economic Indicators

- GDP
Measures the overall economic output of a country.
- Interest Rates
Influence borrowing costs and investment returns.
- Inflation
Affects pricing power and consumer spending.
- Unemployment Rate
Indicates the health of the labor market.

Company-Specific Factors

- Management Quality
Assessing the skills and experience of a company's management team is crucial.
- Corporate Governance
Evaluating a company's corporate governance practices can help assess its ethical standards and risk management.
- Competitive Advantage
Identifying a company's sustainable competitive advantage can help assess its long-term growth potential.

By effectively using these tools, investors can gain a deeper understanding of a company's financial health, industry dynamics, and future prospects, ultimately helping them make informed investment decisions.

Sources and related content

ECONOMIC ANALYSIS

Economic analysis is a crucial tool for understanding the broader economic factors that influence stock market performance. By examining macroeconomic indicators, investors can gain insights into the overall direction of the economy and make informed decisions about investment timing. This is particularly valuable for large and institutional investors with diversified portfolios.

The prices of stocks are always subject to market risk. Therefore, utmost precaution is taken before investing in stocks. The main micro indicators considered for this research are

- GDP of the country
- Inflation Rate
- FDI
- Unemployment Rate

GDP OF INDIA

India's current GDP (Gross Domestic Product) is approximately \$3.94 trillion in 2024, making it the world's fifth-largest economy by nominal GDP. It was \$3.73 trillion at the end of 2023. India's economy has seen consistent growth over recent decades, becoming one of the fastest-growing major economies

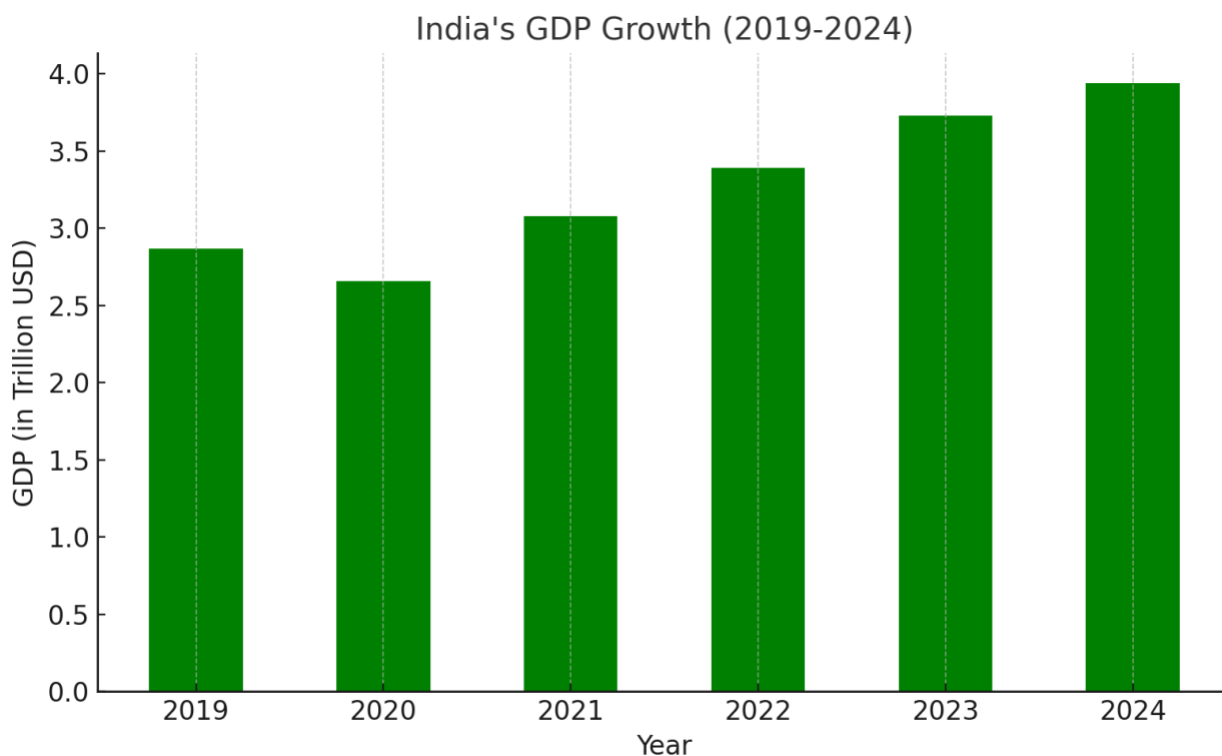
Factors Driving India's Economic Growth:

1. **Economic Reforms:** India has implemented reforms like liberalization and globalization, which have opened up its markets and encouraged economic growth.
2. **Large Population:** With a vast and youthful population, India has a large labor force and consumer base, which boosts domestic demand.
3. **Technological Advancements:** India's progress in technology, especially in the IT and telecommunications sectors, has contributed significantly to its economic growth.
4. **Government Initiatives:** Campaigns like "Make in India" have promoted manufacturing and attracted foreign investments to fuel economic expansion.

Challenges to India's Growth:

1. **Infrastructure Development:** India still faces challenges with its infrastructure, such as inadequate transportation and power systems, which can slow down growth.
2. **Income Inequality:** While the economy is growing, income inequality remains high, with many people still living in poverty.
3. **Unemployment:** Despite economic gains, unemployment, especially among young people, remains a concern.

Overall , India's economic growth has been impressive but still faces challenges like infrastructure gaps, inequality, and unemployment.



INFLATION RATE

Inflation refers to the rise in prices of goods and services over time, while deflation is the opposite when prices decrease. For example, if inflation increases by 4% in 2011, it means that people would have to spend 4% more on the same goods compared to 2010.

Inflation in India:

India's inflation rate has fluctuated over the last decade. While inflation was rising in the early 2010s, it has somewhat moderated in recent years. Despite these fluctuations, India's economy has continued to grow steadily, with increasing GDP and a decrease in national debt.

Current Inflation Trends (as of October 2024):

- **Overall Inflation Rate:** As of June 2024, India's inflation rate was 5.08%, as measured by the Consumer Price Index (CPI). This falls within the Reserve Bank of India's (RBI) target range of 2-6%.
- **Food Inflation:** Food prices have surged, with food inflation reaching 8.66%. Prices of essentials like vegetables, pulses, and cereals have significantly risen.
- **Monetary Policy:** The RBI has kept its repo rate steady at 6.50% to maintain economic stability.

2023 Inflation Overview:

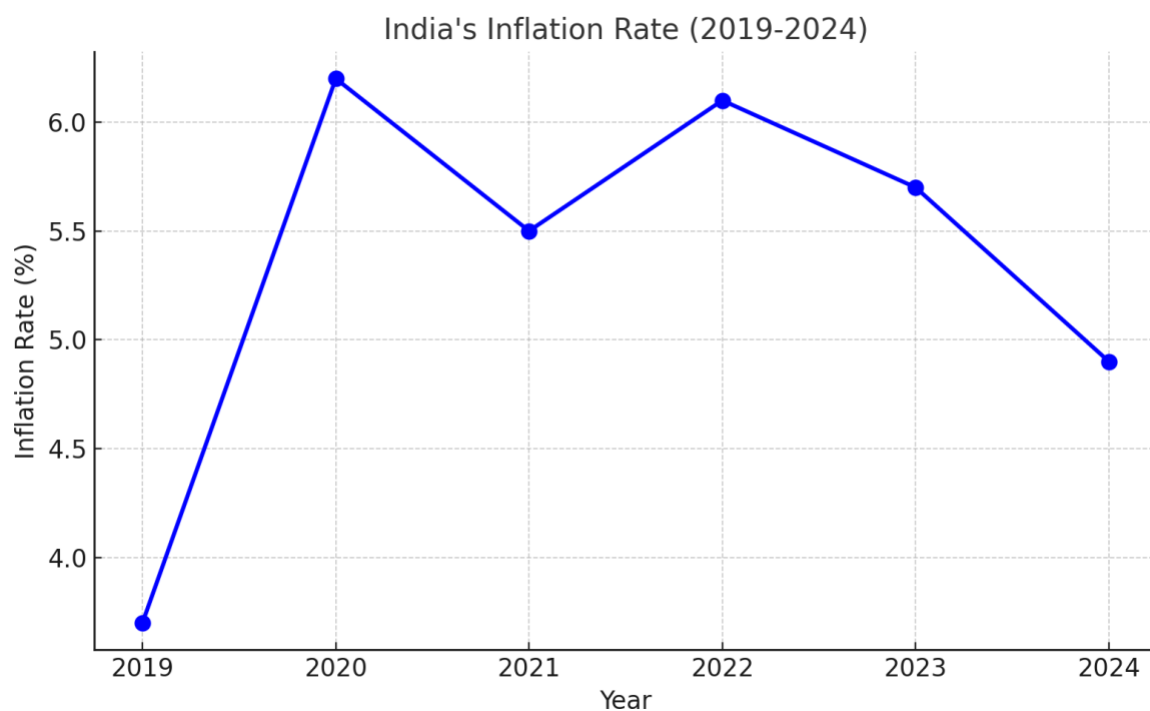
In November 2023, India's inflation rate climbed to 5.55%, marking the first increase in four

months. Food inflation spiked to 8.7%, driven by the impact of a weak monsoon (due to El Niño) on agricultural output. The highest price increases were seen in:

- **Spices:** up 21.6%
- **Pulses:** up 20.2%
- **Vegetables:** up 17.7%, particularly onions and tomatoes
- **Fruits and Cereals:** up 11% and 10.3% respectively

At the same time, the cost of oils and fats dropped by 15%, and fuel and light costs fell slightly by 0.77%. Other sectors like housing, clothing, and footwear saw minor decreases in price growth rates.

Overall, while India's inflation remains moderate overall, food prices continue to experience significant volatility, affecting household expenses.



INDUSTRY ANALYSIS



Industry analysis is a critical component of understanding the dynamics, competitive landscape, and growth potential within a specific sector. For an effective industry analysis, we typically focus on five key areas, which can be adapted based on the industry being studied. Here's a general framework for performing industry analysis:

Examples Specific to the IT Industry:

- **Cybersecurity Infographic:** An infographic showing statistics about cybersecurity threats and preventive measures in the IT industry.
- **Software Development Lifecycle Diagram:** A visual representation of the software development process, including stages like planning, design, coding, testing, and deployment.
- **Cloud Computing Adoption Chart:** A chart showing the growth of cloud computing adoption across various sectors.

Suggestions for Creating These Images:

- **Tools:** Use tools like Canva, Adobe Illustrator, or Microsoft PowerPoint for creating infographics and charts.
- **Data Sources:** Incorporate reliable data from industry reports, market research firms, or government
- **IT Industry Sector Analysis**

1. Industry Overview

- **Definition:** The Information Technology (IT) industry encompasses a wide range of services and products related to computing, software, hardware, telecommunications, and internet-related services.
- **Market Size:** The global IT industry is valued at over \$5 trillion (as of 2024) and continues to grow, driven

by digital transformation across sectors.

- **Key Segments:**

- **Software Development:** Includes applications, enterprise solutions, and software as a service (SaaS).
- **Hardware Manufacturing:** Encompasses computers, servers, and networking equipment.
- **IT Services:** Includes consulting, system integration, and managed services.
- **Telecommunications:** Covers services related to internet and phone services, including mobile and broadband.

2. Key Trends

- **Cloud Computing:** Adoption of cloud services is rapidly increasing, allowing businesses to scale operations and reduce costs.
- **Artificial Intelligence (AI):** AI technologies are being integrated into software solutions, enhancing automation, analytics, and customer service.
- **Cybersecurity:** With increasing cyber threats, companies are investing heavily in cybersecurity solutions to protect their data and infrastructure.
- **Big Data and Analytics:** Organizations are leveraging big data to gain insights and make data-driven decisions, transforming their operational strategies.

3. Competitive Landscape

- **Key Players:** Major companies in the IT industry include:
 - **Tech Giants:** Microsoft, Amazon (AWS), Google, IBM, and Oracle.
 - **Consulting Firms:** Accenture, Deloitte, and Capgemini.
 - **Emerging Startups:** Numerous startups focusing on niche technologies like AI, blockchain, and cybersecurity.
- **Market Share:** Large players dominate the market, but innovation from startups is challenging traditional models.

4. Challenges

- **Talent Shortage:** The IT industry faces a significant talent gap, with high demand for skilled professionals in areas like software development, data science, and cybersecurity.
- **Rapid Technological Changes:** The fast pace of technological advancements requires continuous learning and adaptation from both companies and employees.
- **Cybersecurity Risks:** Increasing cyber threats pose risks to business operations and customer data, necessitating robust security measures.
- **Regulatory Compliance:** Companies must navigate complex regulations regarding data privacy and security, such as GDPR and CCPA.

4. Opportunities

- **Digital Transformation:** Businesses across sectors are investing in digital solutions, creating demand for IT services and products.
- **Emerging Technologies:** Innovations in AI, IoT, and blockchain present new opportunities for IT companies to develop solutions that address evolving market needs.
- **Global Expansion:** Companies can explore international markets, leveraging technology to provide services globally and tap into new customer bases.
- **Sustainability Initiatives:** There is a growing emphasis on sustainable practices, creating opportunities for IT firms to offer eco-friendly solutions.

5. SWOT Analysis

- **Strengths:**
 - Strong growth potential driven by digital transformation.
 - Diverse service offerings across various sectors.
- **Weaknesses:**
 - High competition leading to price pressures.
 - Dependence on skilled labor and talent shortages.
- **Opportunities:**
 - Increased investment in AI and automation.
 - Expanding markets in developing countries.
- **Threats:**
 - Rapid technological changes leading to obsolescence.
 - Cybersecurity threats affecting reputation and customer trust.

6. Future Outlook

- The IT industry is expected to continue its growth trajectory, driven by advancements in technology and increased demand for digital solutions. Companies that can adapt to changing market dynamics and invest in innovation will be well-positioned for success.

Past Performance of the IT Industry



1. 1960s - 1980s: Early Development

- **Mainframe Era:** Dominated by IBM and large organizations using mainframe computers.
- **Software Emergence:** Birth of the software industry with the founding of Microsoft in 1975.

2. 1980s - 1990s: Personal Computing Boom

- **PC Revolution:** Rise of personal computers, notably the IBM PC and Apple Macintosh.
- **Internet Growth:** Commercialization of the internet leading to the dot-com boom.

3. 1990s - Early 2000s: Dot-Com Boom and Bust

- **Rapid Expansion:** Surge in internet-based startups and tech stock prices.
- **Market Correction:** The dot-com bubble burst in 2000, leading to significant failures in the tech sector.

4. 2000s: Recovery and Consolidation

- **New Technologies:** Emergence of mobile computing and cloud services.
- **Mergers & Acquisitions:** Major tech firms acquired startups to enhance capabilities.

5. 2010s: Cloud Computing and Mobile Technologies

- **Cloud Adoption:** Shift towards cloud services, with Amazon Web Services leading the market.
- **Mobile Revolution:** Proliferation of smartphones and apps transformed consumer access to services.

6. 2020s: Digital Transformation

- **COVID-19 Impact:** Accelerated digital transformation and remote work solutions.
- **Increased Cybersecurity Focus:** Significant investment in cybersecurity measures.
- **Emerging Technologies:** Growth in AI, IoT, and blockchain technologies.

Key Performance Indicators (KPIs)

- **Revenue Growth:** CAGR of 5-7% over the past decade, with market size surpassing \$5 trillion in 2024.
- **Employment Growth:** Millions of jobs created, especially in software development and cybersecurity.

Conclusion

The IT industry's past performance reflects resilience, with consistent growth driven by technological advancements and digital transformation. It remains essential for economic development across sectors.

Company Analysis

Definition and Importance

Company analysis involves studying various characteristics of a company to evaluate its operational and financial performance, as well as its future prospects. Once investors decide to invest in a particular industry based on economic and industry analysis, selecting specific companies for investment becomes crucial.

Estimating Future Performance

The estimation of future dividends and earnings relies heavily on a company's past performance and managerial competence. This evaluation is conducted within a broader framework that includes economy-wide and industry-specific analysis. The outcome of a comprehensive company analysis is an expected future cash inflow from the company's shares, which is essential for determining the intrinsic value of those shares.

Intrinsic Value Assessment

The intrinsic value of a share is then compared with its prevailing market price to ascertain whether it is undervalued or overvalued.

- If a company's shares are priced lower than their intrinsic value, they are considered **undervalued**, indicating a potential buying opportunity for investors.
- Conversely, if the shares are priced higher than their intrinsic value, they are deemed **overvalued**, suggesting that investors should refrain from purchasing them. In such cases, existing shareholders may consider selling their shares.

Key Factors in Intrinsic Value

The intrinsic value of a company is influenced by the amount of dividends distributed and its growth rate, both of which are tied to the company's earnings. Therefore, analyzing a company's earnings is critical in the company analysis process.

Data Sources for Company Analysis

Data for company analysis can be sourced from various financial documents, including:

- **Balance Sheet**
- **Income Statement**
- **Cash Flow Statement**
- **Notes to Financial Statements**
- **Auditor's Report**
- **Social and Sustainability Reports** (if applicable)
- **Corporate Governance Reports**

Parameters of Study

Company analysis encompasses several key areas of study, such as:

- Financial ratio analysis, particularly focusing on earnings
- Evaluation of company management and corporate governance
- Analysis of product differentiation and innovation

Purpose of Company Analysis

Investment Decision-Making

Helps investors assess a company's viability before investing.

Determining Intrinsic Value

Estimates the true value of shares for informed buying or selling decisions.

Understanding Financial Health

Analyzes financial statements to gauge liquidity, profitability, and efficiency.

Evaluating Management Performance

Assesses management's effectiveness in achieving strategic goals.

Identifying Risks and Opportunities

Highlights potential risks and growth opportunities.

Comparative Analysis

Compares the company with peers to understand market position and competitive advantages.

Forecasting Future Performance

Projects future earnings and cash flows based on historical data.

Guiding Corporate Strategy

Assists management in refining strategies and making data-driven decisions.

Supporting Valuation Models

Provides data for accurate valuation using various models.

CHAPTER 4

RESEARCH METHODOLOGY

The research methodology adopted for this study is analytical. An attempt has been made to collect data on financials statements of IT Sector Companies for the purpose of fundamental analysis of State TCS and WIPRO.

Primary Data :- Primary data is that which had not been previously generated. It's collected at the time of research by the researcher themselves. We can refer to sources of primary data as primary research methods.

Let's take a look at some examples of sources of primary data collection.

- Experiments can be conducted in laboratories or in real-world (or 'field') settings. The point of experiments is to establish correlation and/or causation between multiple variables.
- Social surveys are a popular method of data collection both within and outside sociology. They involve systematically collecting data from a large sample in order to gather information and establish relationships between multiple variables.

Secondary Data :- Secondary data is a second-hand data that is already collected and recorded by some researchers for their purpose, and not for the current research problem.

- Secondary data is from recorded and published sources. This data is easier to collect, quick and commercial. This data is not new. Secondary data is supportive in character and used only after testing its accuracy.
- There are various types of secondary data such as published printed sources, government records and published electronic, unpublished personal records.
- A set of theoretical data has been collected from various research, papers related to the project and also various websites in the project.

In this project study researcher has taken into account published financial statements and relevant data for the purpose of fundamental analysis.

CHAPTER 5

TECHNICAL ANALYSIS OF IT SECTOR



TECHNICAL ANALYSIS

Technical analysis is like a detective examining clues to predict future price movements in financial markets. Instead of relying solely on financial statements and company fundamentals, technical analysts focus on studying historical price charts and patterns.

By analyzing these charts, they can identify trends, support and resistance levels, and other indicators that may signal potential buying or selling opportunities.

It's a way to understand market sentiment and anticipate how investors might react to different price movements. Think of it as using past price data to create a roadmap for future price action. While it's not a guaranteed crystal ball, technical analysis can provide valuable insights and help you make more informed investment decisions.

Basics of Technical Analysis

Technical analysis is a method of predicting future stock prices based on historical data. It assumes that past price and volume patterns can be used to identify trends and make informed investment decisions. By studying charts, ratios, and patterns, investors can determine the best time to buy or sell a particular stock.

Technical analysis is based on the premise that "history repeats itself" and that past price and volume data can be used to predict future price movements. This is because technical analysts believe that market prices are influenced by a variety of factors, including supply and demand, investor sentiment, and economic conditions. These factors can cause prices to move in predictable patterns over time.

Technical analysts use a variety of tools and techniques to identify trends and patterns in stock prices.

These include:

- **Charts:** Charts are used to visualize price and volume data over time. There are many different types of charts, including line charts, bar charts, and candlestick charts.
- **Indicators:** Indicators are mathematical formulas that are applied to price and volume data to identify trends and patterns. Some common indicators include moving averages, relative strength index (RSI), and Bollinger Bands.
- **Patterns:** Patterns are recurring chart formations that can be used to predict future price movements. Some common patterns include head and shoulders, triangles, and double tops and bottoms.

Technical analysis can be a valuable tool for investors who want to make informed investment decisions. However, it is important to note that technical analysis is not a guaranteed method of predicting future stock prices. There are many factors that can influence stock prices, and it is impossible to predict all of them.

Technical analysis offers several advantages for investors:

- **Objective Analysis**

It relies on historical data and mathematical calculations, providing an objective approach to evaluating investments. This minimizes the influence of emotions and biases that can often cloud judgment.

- **Trend Identification**

By studying charts and indicators, technical analysts can precisely identify potential trends and patterns in stock prices. This can help investors anticipate future price movements and make timely decisions.

- **Timing Entry and Exit Points**

Technical analysis can assist in determining optimal entry and exit points for investments. By identifying support and resistance levels, investors can potentially buy at lower prices and sell at higher prices.

- **Versatility**

Technical analysis can be applied to various financial instruments, including stocks, bonds, commodities, and currencies. It is a versatile tool that can be used in different market conditions.

- **Independence**

Technical analysis is not reliant on fundamental analysis, which involves evaluating a company's financial health and prospects. This independence can provide a different perspective on investment opportunities.

- **Accessibility**

Technical analysis tools and resources are widely available, making it accessible to investors of all levels. There are numerous online platforms and software programs that can assist in analyzing charts and indicators.

- **Price Action Focus**

Technical analysis primarily focuses on price action, which can be a valuable tool for short-term trading or swing trading strategies. It can help identify opportunities for profit based on price movements.

- **Confirmation of Fundamental Analysis**

While technical analysis can be used independently, it can also be used to confirm or contradict fundamental analysis. If technical indicators align with fundamental factors, it can strengthen the investment case.

- **Customization**

Technical analysis can be customized to suit individual investment styles and risk tolerance. Investors can choose different indicators, timeframes, and strategies to align with their preferences.

- **Educational Value**

Learning technical analysis can enhance financial knowledge and understanding of market dynamics. It can provide a valuable foundation for making informed investment decisions.

Technical analysis offers several advantages for investors:

- **Lagging Indicators**

Many technical indicators are based on historical data, which means they may lag behind actual market movements. This can lead to missed opportunities or incorrect signals.

- **Subjectivity**

While technical analysis aims to be objective, there is still an element of subjectivity involved in interpreting charts and indicators. Different analysts may have different interpretations of the same data.

- **False Signals**

Technical analysis can sometimes generate false signals, leading to incorrect buy or sell decisions. This can result in losses for investors.

- **Market Inefficiencies**

Technical analysis assumes that market prices reflect all available information. However, markets

are not always efficient, and unexpected events can occur that disrupt patterns and trends.

- **Overreliance on Past Data**

- Technical analysis relies heavily on past price and volume data. If market conditions change significantly, past patterns may no longer be relevant.

- **Limited Predictive Power**

Technical analysis can help identify trends and patterns, but it does not guarantee future price movements. There is always the possibility of unexpected events that can render technical analysis ineffective.

- **Complexity**

Technical analysis can be complex, requiring a significant amount of time and effort to learn and apply effectively. Investors who are not comfortable with technical concepts may find it challenging to use.

- **Risk of Overtrading**

Technical analysis can encourage frequent trading, which can increase transaction costs and erode profits. Overtrading can also lead to emotional decision-making and impulsive actions.

- **Lack of Fundamental Analysis**

Technical analysis focuses solely on price action and does not consider the underlying fundamentals of a company. This can lead to investments in companies that may be financially unsound or facing significant challenges.

- **Psychological Challenges**

Using technical analysis can be psychologically demanding, as it requires discipline, patience, and the ability to handle losses. Investors who are not emotionally prepared may find it difficult to stick to their trading plan.

Types of Indicators in Technical Analysis:

- **Relative Strength Index (RSI)**

The relative strength index (RSI) is a [technical indicator](#) used in the analysis of [financial markets](#). Traditionally, the Relative Strength Index is considered to signal overbought conditions when above 70 and oversold conditions when under 30. The levels can be adjusted, however, to better fit the price movement of a specific [security](#) a trader is watching. If, for example, a security's RSI consistently hits above the 70 mark or below the 30 mark without correctly forecasting a change in price trend, a trader might adjust the upper end to 80 and/or the lower end to 20 to get more reliable trading signals.

- **Volume**

In technical analysis, volume indicators measure the trading activity or "buzz" behind price movements. By analysing changes in volume alongside price trends, traders can gauge the strength of a move, confirm potential breakouts or reversals, and identify areas of high investor interest, giving them insights to complement their overall trading strategy.

- **ADX**

The ADX indicator, short for Average Directional Movement Index, acts like a trend meter in technical analysis. It shows the strength of a prevailing trend, regardless of its direction (up or down), by measuring the intensity of price movement. Values above 25 typically indicate a strong trend, while those below suggests a sideways market.

- **Donchian Channel**

The Donchian channel, pictured as two bands around a price chart, acts like a volatility meter in technical analysis. The top band marks the highest high over a set period, the bottom band marks the lowest low, and the space between gauges' market turbulence. Breakouts above or below the channel signal potential trends, while price bouncing within the channel hints at consolidation. It's a simple yet versatile tool for traders to identify potential entry and exit points, assess trends, and gauge market sentiment.

- **Exponential Moving Average (EMA)**

The Exponential Moving Average (EMA) is a technical indicator that tracks price trends, but unlike a simple average, it gives greater weight to recent price data, reacting quickly to changes. This makes it helpful for identifying early trend shifts and potential support/resistance levels. Its responsiveness comes with the trade-off of being more prone to false signals, so it's often used alongside other indicators for confirmation.

CHAPTER 6

COMPANY ANALYSIS

5.1 INTRODUCTION

A financial statement is a summary report that outlines the financial activities and position of a business or individual over a specific period. The three main types of financial statements are the income statement, the balance sheet, and the cash flow statement. Here's a brief overview of each:

1. Income Statement (Profit & Loss Statement) –
 - Shows a company's revenues and expenses over a specific period.
 - Net income (or loss) is the key figure, indicating profitability.

2. Balance Sheet –
 - Presents a snapshot of a company's financial position at a specific point in time.
 - Assets (what the company owns), liabilities (what it owes), and equity (the residual interest of the owners) are the three main sections.

3. Cash Flow Statement –
 - Details how changes in balance sheet accounts affect cash and cash equivalents. Operating, investing, and financing activities are reported separately.

These financial statements are essential tools for assessing the financial health and performance of a business. They are typically prepared by accountants based on accurate and up-to-date financial data, allowing stakeholders to make informed decisions about the entity's operations and future prospects.

5.2 FINANCIAL STATEMENTS OF TCS & WIPRO



Tata Consultancy Services (TCS) Tata Consultancy Services (TCS) is one of the largest multinational IT service and consulting companies. It is headquartered in Mumbai, India, but has offices globally. TCS is well known in the e-governance, banking and financial services, telecommunications, education and healthcare markets.

TCS is a major player in the IT space. It currently has over 600,000 employees and a market cap of over \$200 billion.

Tata Consultancy Services was formed in 1968 as one of the first IT companies in India. Many of its early customers were national banks. Its history with the large enterprise financial sector from the early days of computers provided much experience in the financial services market.

In 2004, TCS became a publicly traded company. It has since become one of the largest and most profitable companies in India.

TCS is a subsidiary of Tata Sons, a major Indian holding company. Tata Sons owns shares in many large Indian companies. Some of the industries of these other companies include airlines, automotive, consumer goods, hotels and steel manufacturing.

TCS's strategic vision, coupled with its robust financial performance and global footprint, positions it as a leading player in the IT services industry, driving innovation and delivering value to stakeholders worldwide.

TATA CONSULTANCY SERVICES

INCOME STATEMENT

PROFIT & LOSS A/C OF TATA CONSULTANCY SERVICES (in Rs. Cr.)	Mar-24	Mar-23	Mar-22	Mar-21
INCOME				
REVENUE FROM OPERATIONS [GROSS]	202,359.00	190,354.00	160,341.00	135,963.00
REVENUE FROM OPERATIONS [NET]	202,359.00	190,354.00	160,341.00	135,963.00
TOTAL OPERATING REVENUES	202,359.00	190,354.00	160,341.00	135,963.00
Other Income	7,273.00	5,328.00	7,486.00	5,400.00
TOTAL REVENUE	209,632.00	195,682.00	167,827.00	141,363.00
EXPENSES				
Operating And Direct Expenses	3,347.00	1,416.00	1,010.00	1,230.00
Other Expenses	40,026.00	41,723.00	31,989.00	25,377.00
TOTAL EXPENSES	151,072.00	143,992.00	118,104.00	99,243.00
PROFIT/LOSS BEFORE EXCEPTIONAL, EXTRAORDINARY ITEMS AND TAX	58,560.00	51,690.00	49,723.00	42,120.00
Exceptional Items	-958	0	0	-1,218.00
PROFIT/LOSS BEFORE TAX	57,602.00	51,690.00	49,723.00	40,902.00
TAX EXPENSES-CONTINUED OPERATIONS				
Current Tax	14,178.00	12,946.00	11,931.00	10,300.00
TOTAL TAX EXPENSES	14,043.00	12,584.00	11,536.00	9,942.00
PROFIT/LOSS AFTER TAX AND BEFORE EXTRAORDINARY ITEMS	43,559.00	39,106.00	38,187.00	30,960.00
PROFIT/LOSS FROM CONTINUING OPERATIONS	43,559.00	39,106.00	38,187.00	30,960.00
PROFIT/LOSS FOR THE PERIOD	43,559.00	39,106.00	38,187.00	30,960.00
EARNINGS PER SHARE				
Basic EPS (Rs.)	119.44	106.88	103.24	82.78
Diluted EPS (Rs.)	119.44	106.88	103.24	82.78
DIVIDEND AND DIVIDEND PERCENTAGE				
Equity Share Dividend	25,137.00	41,347.00	13,317.00	10,850.00
Tax On Dividend	0	0	0	0
Equity Dividend Rate (%)	7,300.00	11,500.00	4,300.00	3,800.00

SOURCE - <https://www.moneycontrol.com/india/>

BALANCE SHEET

BALANCE SHEET OF TATA CONSULTANCY SERVICES (in Rs. Cr.)	Mar-24	Mar-23	Mar-22	Mar-21
EQUITIES AND LIABILITIES				
SHAREHOLDER'S FUNDS				
Equity Share Capital	362	366	366	370
TOTAL SHARE CAPITAL	362	366	366	370
Reserves and Surplus	71,758.00	74,172.00	76,807.00	74,424.00
TOTAL RESERVES AND SURPLUS	71,758.00	74,172.00	76,807.00	74,424.00
TOTAL SHAREHOLDERS FUNDS	72,120.00	74,538.00	77,173.00	74,794.00
NON-CURRENT LIABILITIES				
Long Term Borrowings	0	0	0	0
Deferred Tax Liabilities [Net]	154	190	129	365
Other Long Term Liabilities	5,813.00	5,775.00	6,060.00	5,697.00
Long Term Provisions	0	0	0	0
TOTAL NON-CURRENT LIABILITIES	5,967.00	5,965.00	6,189.00	6,062.00
CURRENT LIABILITIES				
Short Term Borrowings	0	0	0	0
Trade Payables	14,599.00	13,768.00	10,082.00	7,962.00
Other Current Liabilities	28,391.00	25,277.00	26,442.00	19,213.00
Short Term Provisions	71	279	1,377.00	1,350.00
TOTAL CURRENT LIABILITIES	43,061.00	39,324.00	37,901.00	28,525.00
TOTAL CAPITAL AND LIABILITIES	121,148.00	119,827.00	121,263.00	109,381.00
NON-CURRENT ASSETS				
Tangible Assets	14,490.00	14,881.00	15,506.00	15,697.00
Intangible Assets	463	809	1,018.00	362
Capital Work-In-Progress	1,450.00	1,103.00	1,146.00	861
Other Assets	0	0	0	0
FIXED ASSETS	16,403.00	16,793.00	17,670.00	16,920.00
Non-Current Investments	2,405.00	2,405.00	2,405.00	2,405.00
Deferred Tax Assets [Net]	2,524.00	2,464.00	2,779.00	3,160.00
Long Term LoansAnd Advances	2	3	8	2
Other Non-Current Assets	4,896.00	5,378.00	4,209.00	3,734.00
TOTAL NON-CURRENT ASSETS	26,230.00	27,043.00	27,071.00	26,221.00
CURRENT ASSETS				
Current Investments	29,840.00	35,738.00	29,262.00	28,324.00
Inventories	27	27	19	7
Trade Receivables	46,068.00	42,798.00	29,852.00	25,222.00
Cash And Cash Equivalents	6,599.00	4,543.00	13,692.00	3,142.00
Short Term Loans Advances	317	332	5,653.00	10,486.00
Other Current Assets	12,067.00	9,346.00	15,714.00	15,979.00
TOTAL CURRENT ASSETS	94,918.00	92,784.00	94,192.00	83,160.00
TOTAL ASSETS	121,148.00	119,827.00	121,263.00	109,381.00

SOURCE - <https://www.moneycontrol.com/india/>

CASH FLOW

CASH FLOW OF WIPRO (in Rs. Cr.)	Mar-24	Mar-23	Mar-22	Mar-21
NET PROFIT/LOSS BEFORE EXTRAORDINARY ITEMS AND TAX	9,118.60	9,176.70	12,135.30	10,060.90
Net Cash-Flow From Operating Activities	14,215.90	11,191.60	7,240.50	12,727.70
Net Cash Used In Investing Activities	2,237.00	-4,761.60	-12,611.10	-1,275.00
Net Cash Used From Financing Activities	-17,158.70	-6,803.60	494.70	-12,105.70
Foreign Exchange Gains / Losses	-30.60	2.50	-9.20	-7.80
Net Inc/Dec In Cash And Cash Equivalents	-736.40	-371.10	-4,885.10	-660.80
Cash And Cash Equivalents Begin of Year	4,527.00	4,898.10	9,783.20	10,444.00
Cash And Cash Equivalents End Of Year	3,790.60	4,527.00	4,898.10	9,783.20

SOURCE - <https://www.moneycontrol.com/india/>

FUTURE PLANS OF TATA CONSULTANCY SERVICES

Tata Consultancy Services (TCS) has ambitious plans for the future. They aim to double their revenue to \$50 billion by 2030 through an organizational overhaul. This will involve creating four specialized internal teams focusing on business transformation, incubation, enterprise growth, and new business models. These changes are designed to better meet the needs of their diverse client base, from startups to large global corporations.

In addition, TCS is expanding its global footprint by establishing multiple innovation centers in the U.S. and Europe. They are also committed to advancing in areas such as cloud services, data analytics, and artificial intelligence, which are expected to drive significant growth. Recently, TCS has been selected for various high-profile projects, including digital transformation initiatives for major companies like Bombardier and Bayer.

These strategic moves and investments underline TCS's focus on staying at the forefront of technological innovation and maintaining strong growth in the evolving digital landscape.



Western India Palm Refined Oils Limited (Wipro)

WIPRO is a multinational corporation that provides information technology, consulting, and business process services. It was founded in 1945 as Western India Vegetable Products Limited, a manufacturer of vegetable and refined oils, and has undergone several name changes since then. The current name, WIPRO, is not an acronym and does not stand for anything. It is simply the shortened version of the original name of the company. Over the years, Wipro has diversified its business and expanded into various sectors, including information technology, consulting, and business process services. It has also made several acquisitions to grow its business, including the acquisition of Spectramind (a business process outsourcing company) in 2000, New Logic (a semiconductor design company) in 2002, and Infocrossing (a provider of managed services) in 2007.

Today, Wipro is a global company with a presence in over 50 countries and a workforce of over 160,000 employees. It is headquartered in Bangalore, India, and is listed on the Bombay Stock Exchange and the National Stock Exchange of India.

MISSION : Wipro is an Indian multinational corporation that provides information technology, consulting, and business process services. Here is Wipro's mission statement, as stated on the company's website:

"Our mission is to help our clients achieve their business objectives through a combination of process excellence, quality frameworks and service delivery innovation."

In other words, Wipro's mission is to provide high-quality services that help its clients improve their businesses and achieve their goals.

VISION : Wipro Limited is a global information technology, consulting and business process services company. According to Wipro's website, the company's vision is to "be a globally respected corporation that provides best-in-class products, solutions and services, and to be the most trusted partner for its stakeholders."

Wipro provides a range of IT services to its clients, including application development, maintenance, and support; infrastructure management; testing services; and digital transformation services.

Western India Palm Refined Oils Limited (Wipro)

INCOME STATEMENT

	Mar-24	Mar-23	Mar-22	Mar-21
INCOME				
REVENUE FROM OPERATIONS [GROSS]	66,792.40	67,753.40	59,574.40	50,299.40
REVENUE FROM OPERATIONS [NET]	66,792.40	67,753.40	59,574.40	50,299.40
TOTAL OPERATING REVENUES	66,792.40	67,753.40	59,574.40	50,299.40
Other Income	3,045.80	2,354.20	4,706.10	2,382.90
TOTAL REVENUE	69,838.20	70,107.60	64,280.50	52,682.30
EXPENSES				
Operating And Direct Expenses	13,148.50	13,918.90	14,109.60	10,675.40
Other Expenses	3,389.10	4,122.50	1,028.80	480.5
TOTAL EXPENSES	57,420.70	57,838.70	49,016.30	39,997.50
PROFIT/LOSS BEFORE EXCEPTIONAL, EXTRAORDINARY ITEMS AND TAX	12,417.50	12,268.90	15,264.20	12,684.80
Exceptional Items	0	0	0	0
PROFIT/LOSS BEFORE TAX	12,417.50	12,268.90	15,264.20	12,684.80
TAX EXPENSES-CONTINUED OPERATIONS				
Current Tax	3,148.50	2,740.50	3,194.10	2,243.00
Less: MAT Credit Entitlement	0	0	0	0
Deferred Tax	150.4	351.7	-65.2	380.9
Tax For Earlier Years	0	0	0	0
TOTAL TAX EXPENSES	3,298.90	3,092.20	3,128.90	2,623.90
PROFIT/LOSS AFTER TAX AND BEFORE EXTRAORDINARY ITEMS	9,118.60	9,176.70	12,135.30	10,060.90
PROFIT/LOSS FROM CONTINUING OPERATIONS	9,118.60	9,176.70	12,135.30	10,060.90
PROFIT/LOSS FOR THE PERIOD	9,118.60	9,176.70	12,135.30	10,060.90
EARNINGS PER SHARE				
Basic EPS (Rs.)	17.24	16.75	22.2	17.81
Diluted EPS (Rs.)	17.19	16.75	22.14	17.77
DIVIDEND AND DIVIDEND PERCENTAGE				
Equity Share Dividend	522.4	548.7	3,289.10	547.8
Tax On Dividend	0	0	0	0
Equity Dividend Rate (%)	50	50	300	50

SOURCE - <https://www.moneycontrol.com/india/>

BALANCE SHEET

BALANCE SHEET OF WIPRO (in Rs. Cr.)	Mar-24	Mar-23	Mar-22	Mar-21
EQUITIES AND LIABILITIES				
SHAREHOLDER'S FUNDS				
Equity Share Capital	1,045.00	1,097.60	1,096.40	1,095.80
TOTAL SHARE CAPITAL	1,045.00	1,097.60	1,096.40	1,095.80
Reserves and Surplus	56,736.90	61,101.50	53,254.30	44,145.80
TOTAL RESERVES AND SURPLUS	56,736.90	61,101.50	53,254.30	44,145.80
TOTAL SHAREHOLDERS FUNDS	57,781.90	62,762.30	54,350.70	45,241.60
NON-CURRENT LIABILITIES				
Deferred Tax Liabilities [Net]	0	0	5.7	14.1
Deferred Tax Liabilities [Net]	448.8	253.1	0	130.5
Other Long Term Liabilities	4,856.40	3,394.50	2,788.60	2,129.20
Long Term Provisions	116.1	54.9	64.1	88.5
TOTAL NON-CURRENT LIABILITIES	5,421.30	3,702.50	2,858.40	2,362.30
CURRENT LIABILITIES				
Short Term Borrowings	4,175.00	5,180.70	7,673.40	5,791.20
Trade Payables	5,839.40	5,300.20	4,685.10	4,348.50
Other Current Liabilities	7,100.30	7,003.90	9,446.90	6,705.30
Short Term Provisions	1,330.70	1,358.00	1,368.30	1,287.40
TOTAL CURRENT LIABILITIES	18,445.40	18,842.80	23,173.70	18,132.40
TOTAL CAPITAL AND LIABILITIES	81,648.60	85,307.60	80,382.80	65,736.30
ASSETS				
NON-CURRENT ASSETS				
Tangible Assets	7,297.80	8,233.80	7,386.60	6,578.70
Intangible Assets	561.7	590.9	651.1	709.4
Capital Work-In-Progress	669.7	603.8	1,584.50	1,848.00
Fixed Assets	8,529.20	9,428.50	9,622.20	9,136.10
Non-Current Investments	20,680.60	19,372.80	16,557.20	8,206.70
Deferred Tax Assets [Net]	25.10	6.80	53.30	47.40
Long Term Loans And Advances	0	0	0	0
Other Non-Current Assets	1,849.90	2,461.70	2,377.90	2,966.60
Total Non-Current Assets	31,084.80	31,329.80	28,610.60	20,356.80
CURRENT ASSETS				
Current Investments	30,143.70	29,712.60	24,073.70	17,495.20
Inventories	72.90	91.30	87.50	91.00
Trade Receivables	11,648.40	13,273.20	9,295.40	8,046.20
Cash And Cash Equivalents	3,790.60	4,527.00	4,898.10	9,783.20
Short Term Loans and Advances	0.00	1,232.60	1,913.00	4,201.50
Other Current Assets	4,908.20	5,141.10	11,504.50	5,762.40
Total Current Assets	50,563.80	53,977.80	51,772.20	45,379.50
Total Assets	81,648.60	85,307.60	80,382.80	65,736.30

SOURCE - <https://www.moneycontrol.com/india/>

CASH FLOW

CASH FLOW OF WIPRO (in Rs. Cr.)	Mar-24	Mar-23	Mar-22	Mar-21
NET PROFIT/LOSS BEFORE EXTRAORDINARY ITEMS AND TAX	24324	23268	21235	18048
Net Cash-Flow From Operating Activities	20787	19169	22096	19902X
Net Cash Used In Investing Activities	-3261	821	-3150	-6309
Net Cash Used From Financing Activities	-15825	-25857	-24275	-9566
Foreign Exchange Gains / Losses	-44	131	-13	23
Net Inc/Dec In Cash And Cash Equivalents	1657	-5736	-5342	4050
Cash And Cash Equivalents Begin of Year	6534	12270	17612	13562
Cash And Cash Equivalents End Of Year	8191	6534	12270	17612

SOURCE - <https://www.moneycontrol.com/india/>



Infosys: A Global Leader in Technology Services

Infosys Limited is a multinational information technology (IT) consulting and services company headquartered in Bengaluru, India. Founded in 1981, Infosys has grown to become one of the world's leading IT service providers, offering a wide range of services to clients across various industries.

Key Strengths and Achievements:

- **Global Presence:** Infosys has a strong global presence with operations in over 50 countries, serving clients in various industries and geographies.
- **Innovation:** The company is known for its focus on innovation and has been a pioneer in adopting emerging technologies such as artificial intelligence, cloud computing, and blockchain.
- **Client Focus:** Infosys has a strong client-centric approach, prioritizing customer satisfaction and delivering high-quality services.
- **Talent Development:** The company invests heavily in talent development and has a reputation for nurturing a skilled and motivated workforce.
- **Corporate Social Responsibility:** Infosys is committed to corporate social responsibility and has undertaken various initiatives in areas such as education, healthcare, and environmental sustainability.

Recent Developments:

- **Digital Transformation:** Infosys has been actively helping clients navigate the digital transformation landscape, providing solutions to modernize their businesses and improve operational efficiency.
- **Cloud Services:** The company has expanded its cloud services offerings, providing comprehensive solutions for cloud migration, management, and optimization.
- **AI and Automation:** Infosys has been leveraging artificial intelligence and automation to enhance its service offerings and drive efficiency.
- **Partnerships and Acquisitions:** The company has formed strategic partnerships and acquired companies to strengthen its capabilities and expand its market reach.

Challenges and Opportunities:

- **Competition:** The IT services industry is highly competitive, with numerous global players vying for market share. Infosys needs to continue to differentiate itself through innovation and quality.
- **Economic Uncertainty:** Economic downturns can impact the demand for IT services. Infosys needs to be resilient and adapt to changing market conditions.
- **Talent Acquisition and Retention:** Attracting and retaining top talent is a critical challenge for the IT industry. Infosys needs to invest in talent development and create a positive work environment.

CONCLUSION, Infosys is well-positioned to capitalize on the growing demand for IT services and digital transformation. With its strong global presence, focus on innovation, and commitment to client satisfaction,

the company is poised for continued success.

INFOSYS

INCOME STATEMENT

	Mar-24	Mar-23	Mar-22	Mar-21
INCOME				
REVENUE FROM OPERATIONS [GROSS]	128933	124014	103940	85912
REVENUE FROM OPERATIONS [NET]	128933	124014	103940	85912
TOTAL OPERATING REVENUES	128933	124014	103940	85912
Other Income	7417	3859	3224	2467
TOTAL REVENUE	136350	127873	107164	88379
EXPENSES				
Operating And Direct Expenses	28449	27275	21958	13533
Changes In Inventories Of FG,WIP And Stock-In Trade	0	0	0	0
Employee Benefit Expenses	65139	62764	51664	45179
Finance Costs	277	157	128	126
Depreciation And Amortisation Expenses	2944	2753	2429	2321
Other Expenses	3588	3281	2490	2743
TOTAL EXPENSES	100397	96230	78669	6429
PROFIT/LOSS BEFORE EXCEPTIONAL, EXTRAORDINARY ITEMS AND TAX	35953	31643	28495	24477
Exceptional Items	0	0	0	0
PROFIT/LOSS BEFORE TAX	35953	31643	28495	24477
TAX EXPENSES-CONTINUED OPERATIONS				
Current Tax	7306	8167	6960	6013
Less: MAT Credit Entitlement	0	0	0	0
Deferred Tax	1413	208	300	416
Tax For Earlier Years	0	0	0	0
TOTAL TAX EXPENSES	8719	8375	7260	6429
PROFIT/LOSS AFTER TAX AND BEFORE EXTRAORDINARY ITEMS	27234	23268	21235	18048
PROFIT/LOSS FROM CONTINUING OPERATIONS	27234	23268	21235	18048
PROFIT/LOSS FOR THE PERIOD	27234	23268	21235	18048
EARNINGS PER SHARE				
Basic EPS (Rs.)	65.62	55.48	50.27	42.37
Diluted EPS (Rs.)	65.56	55.42	50.21	42.33
DIVIDEND AND DIVIDEND PERCENTAGE				
Equity Share Dividend	14733	13675	12700	9158
Tax On Dividend	0	0	0	0
Equity Dividend Rate (%)	920	680	620	540

BALANCE SHEET

BALANCE SHEET OF WIPRO (in Rs. Cr.)	Mar-24	Mar-23	Mar-22	Mar-21
EQUITIES AND LIABILITIES				
SHAREHOLDER'S FUNDS				
Equity Share Capital	2,075	2074	2103	2130
TOTAL SHARE CAPITAL	2075	2074	2103	2130
Reserves and Surplus	79101	64793	66597	69029
TOTAL RESERVES AND SURPLUS	79101	64793	66597	69029
TOTAL SHAREHOLDERS FUNDS	81176	67745	69306	71 531
NON-CURRENT LIABILITIES				
Deferred Tax Liabilities [Net]	1509	866	841	511
Other Long Term Liabilities	5179	5284	4264	4275
Long Term Provisions	0	0	0	0
TOTAL NON-CURRENT LIABILITIES	6688	6150	5105	4786
CURRENT LIABILITIES				
Short Term Borrowings	0	0	0	0
Trade Payables	2493	2426	2669	1562
Other Current Liabilities	23129	23853	21387	15399
Short Term Provisions	1464	1163	920	661
TOTAL CURRENT LIABILITIES	27086	27442	27976	17622
TOTAL CAPITAL AND LIABILITIES	114950	101337	99387	93939
ASSETS				
NON-CURRENT ASSETS				
Tangible Assets	14116	15217	14695	14365
Intangible Assets	211	214	243	234
Capital Work-In-Progress	277	275	411	906
Fixed Assets	14606	15706	15349	115505
Non-Current Investments	23352	23686	22869	22118
Deferred Tax Assets [Net]	0	779	970	955
Long Term Loans And Advances	34	39	34	30
Other Non-Current Assets	6008	9045	7728	7049
Total Non-Current Assets	43998	49225	46950	46657
CURRENT ASSETS				
Current Investments	11307	4476	5467	2037
Inventories	0	0	0	0
Trade Receivables	25152	20773	18966	16394
Cash And Cash Equivalents	8191	6534	13270	17612
Short Term Loans and Advances	208	291	219	229
Other Current Assets	26094	20008	15515	12010
Total Current Assets	70952	52082	52437	48282
Total Assets	114950	101337	99387	93939

CASH FLOW

CASH FLOW OF WIPRO (in Rs. Cr.)	Mar-24	Mar-23	Mar-22	Mar-21
NET PROFIT/LOSS BEFORE EXTRAORDINARY ITEMS AND TAX	24324	23268	21235	18048
Net Cash-Flow From Operating Activities	20787	19169	22096	19902X
Net Cash Used In Investing Activities	-3261	821	-3150	-6309
Net Cash Used From Financing Activities	-15825	-25857	-24275	-9566
Foreign Exchange Gains / Losses	-44	131	-13	23
Net Inc/Dec In Cash And Cash Equivalents	1657	-5736	-5342	4050
Cash And Cash Equivalents Begin of Year	6534	12270	17612	13562
Cash And Cash Equivalents End Of Year	8191	6534	12270	17612

5.3 ANALYSIS AND INTERPRETATION

Ratio helps to summarize large amounts of financial data and make qualitative judgements about the company's financial performance. It is relevant for evaluating a company's performance with respect to liquidity positions, long-term solvency, operational efficiency, overall profitability, inter- firm comparison and trend analysis.

- EARNING PER SHARE** : Earning per share are the portion of the profit which is allocated to each share of the company. It is calculated by dividing the total profit of the company by the total outstanding shares of the company.

16.63

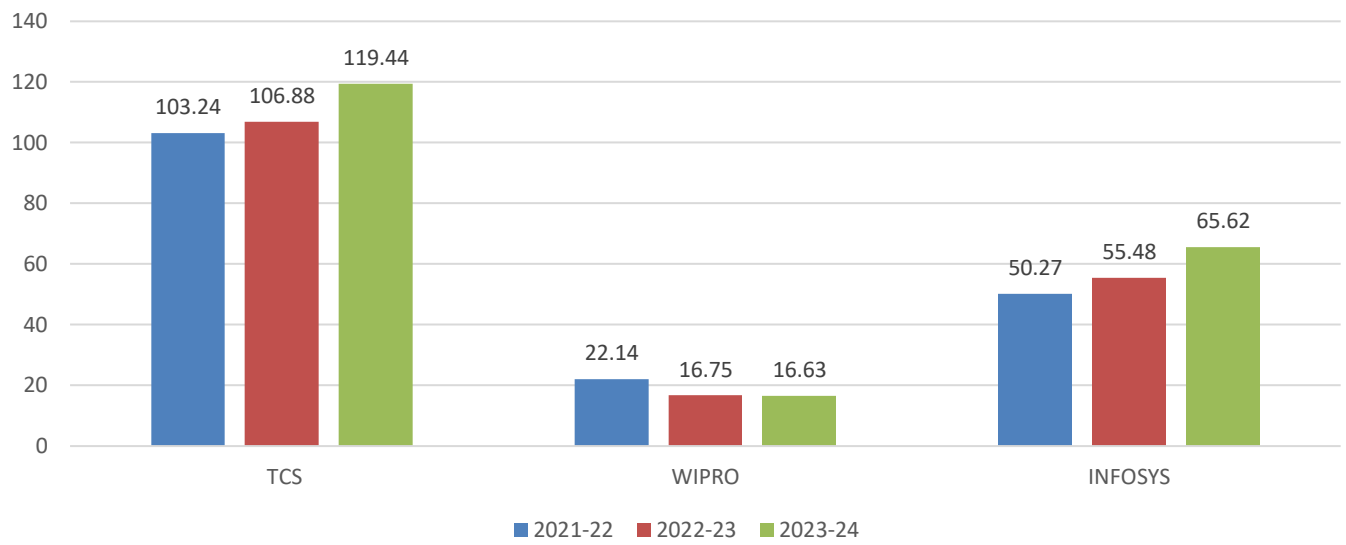
EARNING PER SHARE = $\frac{\text{PROFIT AFTER TAX} - \text{PREFERENCE DIVIDEND}}{\text{NUMBER OF SHARES}}$

Data of EPS for last 3 years

	2021-22	2022-23	2023-24
TCS	103.24	106.88	119.44
WIPRO	22.14	16.75	16.63
INFOSYS	50.27	55.48	65.62

SOURCE - <https://www.moneycontrol.com/india/www.screener.com>

EARNING PER SHARE



CURRENT YEAR EPS OF TCS

(2024) PAT -

43,599

Preference dividend – 0

No. of shares – 362

$$\begin{aligned}\text{EPS} &= \frac{\text{PAT} - \text{Preference Dividend}}{\text{No. of shares}} \\ &= \frac{43,599}{362} \\ &= 119.44\end{aligned}$$

CURRENT YEAR EPS OF WIPRO

(as on Jan 2024)

PAT – 9,118.60

Preference dividend – 0

No. of shares – 548.7

$$\begin{aligned}\text{EPS} &= \frac{\text{PAT} - \text{Preference Dividend}}{\text{No. of shares}} \\ &= \frac{9,119}{549} \\ &= 16.63\end{aligned}$$

CURRENT YEAR EPS OF INFOSYS

(as on Jan 2024)

Net income –

27,234

Preference dividend – 0

No. of shares – 415.29

$$\begin{aligned}\text{EPS} &= \frac{\text{Net Income}}{\text{No. of shares}} \\ &= \frac{27,234}{415.3} \\ &= 65.62\end{aligned}$$

Comparison of EPS Ratio

COMPANIES	PAT – PREFERENCE DIVIDEND(cro re)	NO. OF SHARES	EPS
TCS	43,599	362	119.4
WIPRO	9,119	549	16.63
INFOSYS	27,234	415.29	65.62

INTERPRETATION

- **TCS:** Consistently higher EPS over the past three years, indicating superior profitability.
- **Wipro:** Positive EPS growth, but lower than TCS.
- **Infosys :** Consistently higher EPS over the past three years, indicating superior profitability.

- **Comparison:** TCS has consistently reported higher EPS than Wipro throughout the period, suggesting stronger overall profitability.

Overall: Both companies have shown improved profitability, but TCS has outperformed Wipro in terms of EPS.

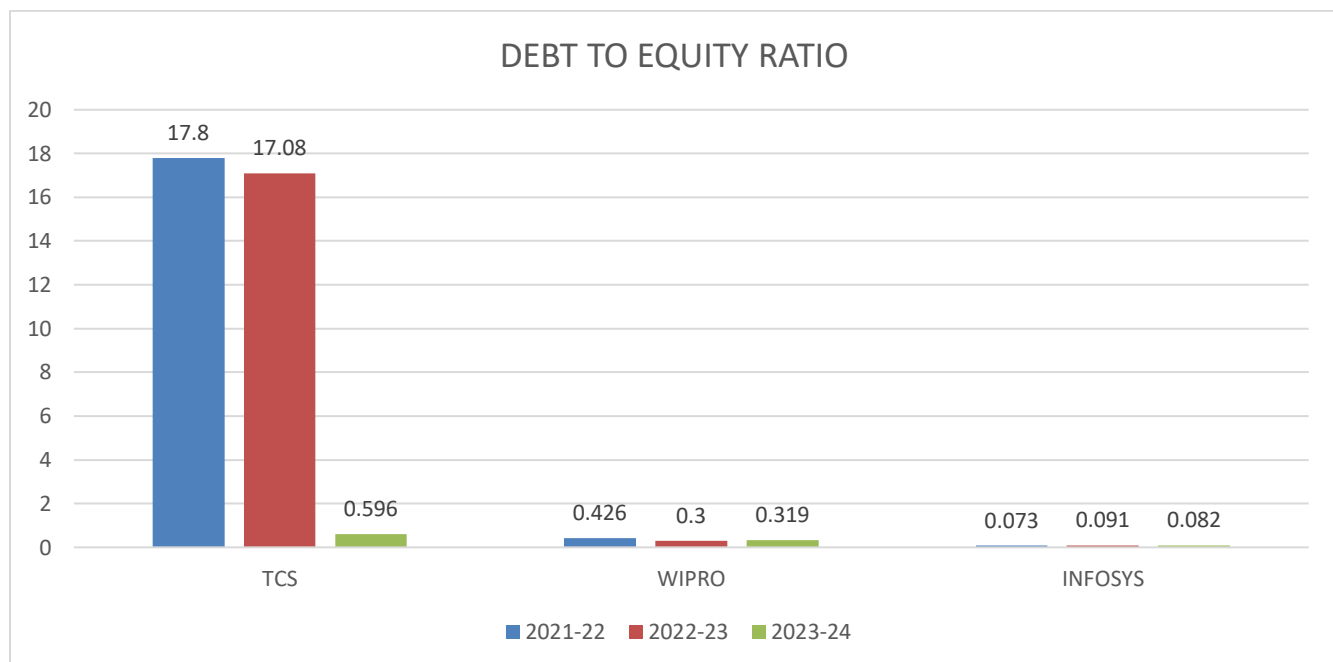
2. **DEBT TO EQUITY RATIO** - A debt to equity ratio is determined to ascertain the soundness of the long-term financial policies of the company. The ratio indicates the extent to which the firm depends upon outsiders for its existence.

$$\text{DEBT-EQUITY RATIO} = \frac{\text{TOTAL LONG TERM DEBTS}}{\text{TOTAL LONG TERM FUNDS}}$$

Data of Debt to Equity Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	17.80	17.08	0.596
WIPRO	0.426	0.300	0.319
INFOSYS	0.073	0.091	0.082

SOURCE - <https://www.moneycontrol.com/india/screner.com>



CURRENT YEAR debt equity OF TCS

(2024) total

liabilities(crore)

43,061

Total shareholder's

equity – 72,120

$$\begin{aligned}\text{Debt to equity} &= \frac{\text{Total Liabilities}}{\text{No. of shareholders' equity}} \\ &= \frac{43,061}{72,120} \\ &= 0.596\end{aligned}$$

CURRENT YEAR EPS OF WIPRO

(2024)total

liabilities –

18,445.40

TOTAL shareholders of

equity – 57,781.90

$$\begin{aligned}\text{Debt to equity} &= \frac{\text{total liabilities}}{\text{No. of shareholder's equity}} \\ &= \frac{18,445}{57,782}\end{aligned}$$

CURRENT YEAR EPS OF INFOSYS

(2024)total

liabilities –

6,688

TOTAL

shareholders of

equity –

81,176

Debt to equity = total liabilities

No. of shareholder's equity

= 6,688

81,176

= 0.082

COMPANIES	TOTAL DEBT CR	TOTAL CR	ROA (%)
TCS	43,061	72,120	0.596
WIPRO	18,445	57,782	0.319
INFOSYS	6,688	81,176	0.082

INTERPRETATION

TCS: With a D/E ratio of 0.596, TCS has a moderate level of leverage. This means that for every rupee of equity, the company has 0.596 paise of debt. While not overly leveraged, TCS's debt-to-equity ratio is slightly higher than Wipro's.

Wipro: With a D/E ratio of 0.319, Wipro has a lower level of leverage compared to TCS. This indicates that Wipro relies less on debt financing and has a stronger financial cushion.
growth opportunities may take on more debt to fund their expansion plans.

Infosys : With a D/E ratio of 0.082, Infosys has a lower level of leverage compared to TCS & Wipro. This indicates that Infosys relies less on debt financing and has a stronger financial cushion.

growth opportunities may take on more debt to fund their expansion plans.

Growth Strategy: Companies may choose to increase their leverage to fund growth initiatives or acquisitions.

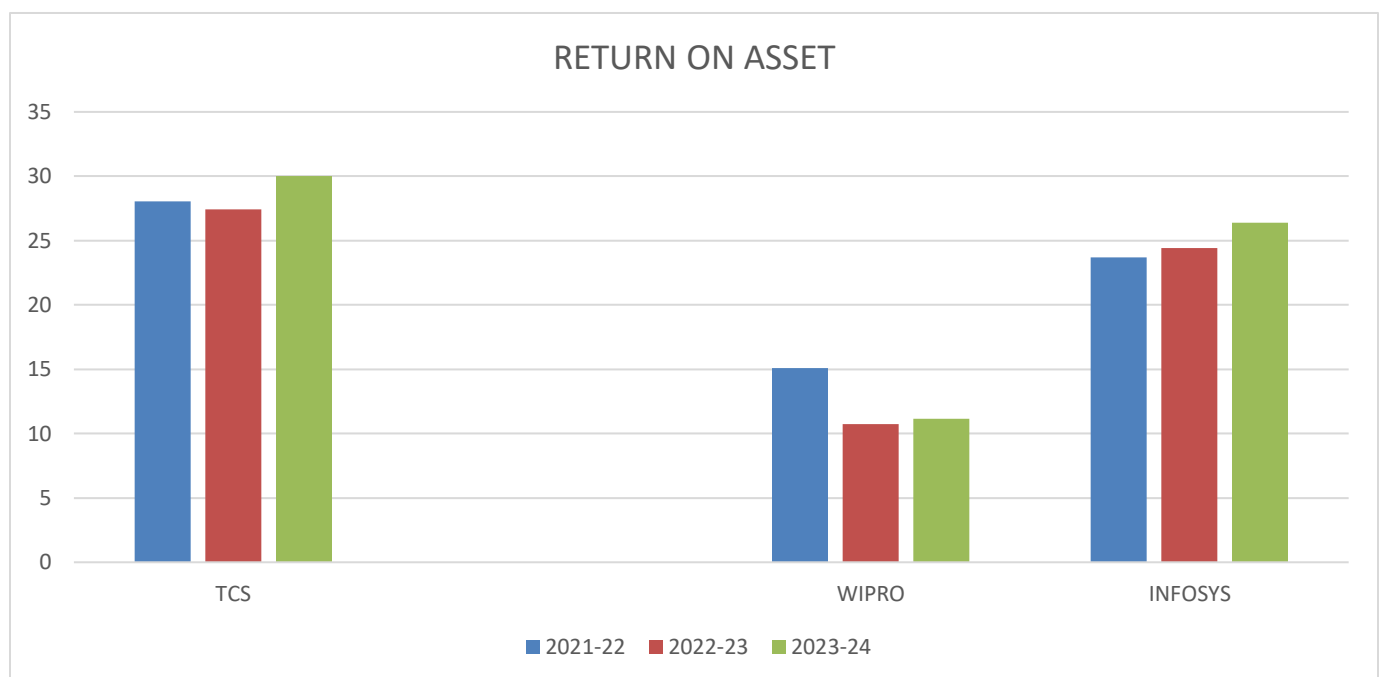
3. **RETURN ON ASSETS (ROA)** - Return on assets (ROA) is a financial metric that measures the profitability of a company in relation to its total assets. It is a key indicator of how efficiently a company is utilizing its assets to generate profits

$$\text{RETURN ON ASSETS} = \frac{\text{NET PROFIT}}{\text{TOTAL ASSETS}}$$

Data of ROA for last 3 years

	2021-22	2022-23	2023-24
TCS	28.06	27.45	30.03
WIPRO	15.09	10.75	11.16
INFOSYS	23.67	24.43	26.41

SOURCE - <https://www.moneycontrol.com/india/wwww.screener.com>



CURRENT ROA OF TCS

Net profit – 43,559 Cr

Total Assets – 145,050

Cr

$$\text{ROA} = \frac{\text{Net Profit}}{\text{Total Assets}}$$

$$= \frac{43,559}{145,050}$$

$$= 30.03\% \text{ }$$

CURRENT ROA OF WIPRO

Net profit – 81,648.60

Total Assets – 9,118.60

$$\text{ROA} = \frac{\text{Net Profit}}{\text{Total Assets}}$$

$$= \frac{81,648.60}{9,118.60}$$

$$= 11.16\% \text{ }$$

CURRENT ROA OF INFOSYS

Net profit – 27,234

Total Assets – 103,143.5

$$\text{ROA} = \frac{\text{Net Profit}}{\text{Total Assets}}$$

$$= \frac{27,234}{103,143.5}$$

$$= 26.41 \%$$

Comparison of ROA

COMPANIES	NET PROFIT CR	TOTAL ASSETS CR	ROA (%)
TCS	43,559	145,050	30.03
WIPRO	18,648.60	9,118.60	11.16
INFOSYS	27,234	103,143.5	26.41

INTERPRETATION

- **TCS:** With an ROA of 30.03%, TCS has demonstrated superior efficiency in utilizing its assets to generate profits. This suggests that TCS is effectively managing its resources and maximizing returns on its investments.
- **Wipro:** While Wipro's ROA of 11.16% is still respectable, it is significantly lower than TCS's. This indicates that Wipro may be less efficient in asset utilization or facing challenges in generating returns from its investments.
- **Infosys:** With an ROA of 26.41 %, Infosys has demonstrated superior efficiency in utilizing its assets to generate profits. This suggests that TCS & Infosys is effectively managing its resources and maximizing returns on its investments

Factors Affecting ROA:

- **Operating Efficiency:** Companies with higher operating efficiency tend to have higher ROAs.
- **Asset Turnover:** A high asset turnover ratio (sales/total assets) can contribute to a higher ROA.
- **Profit Margins:** Higher profit margins can also boost ROA.

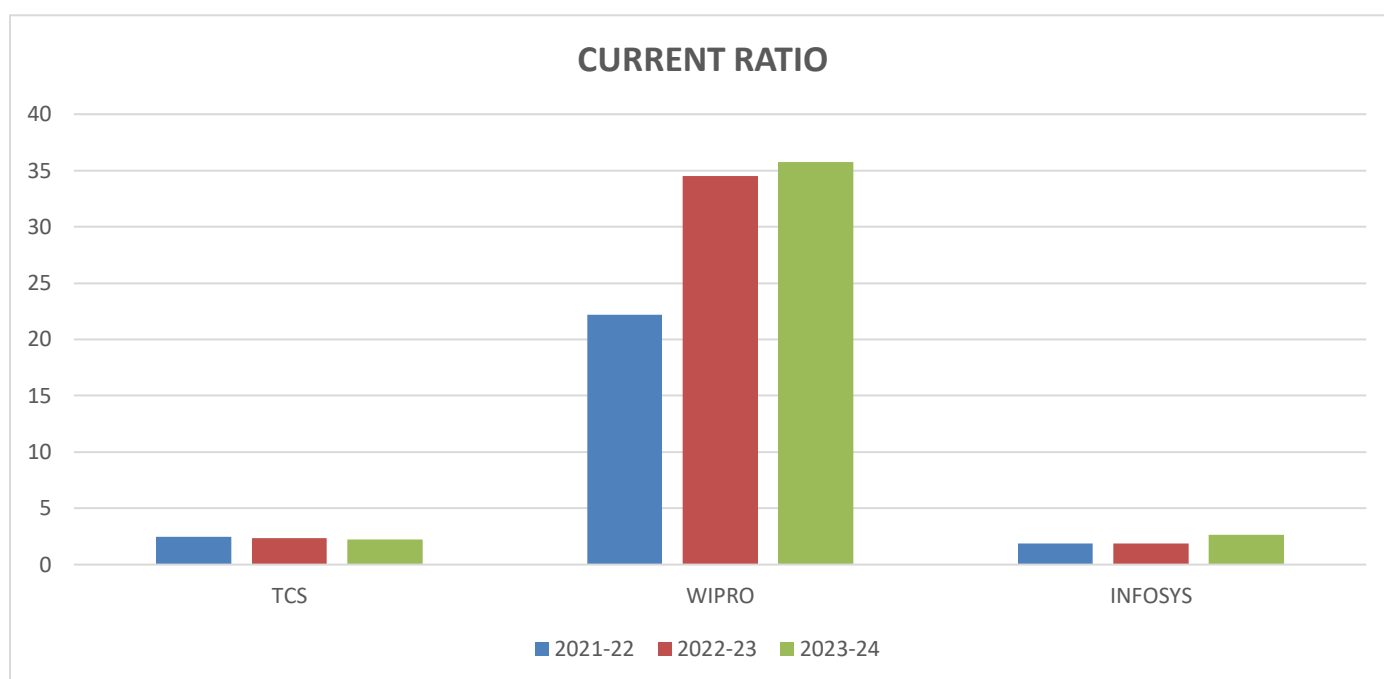
4. **CURRENT RATIO** : This ratio is calculated by dividing the current assets of the company by current liabilities. Current ratio indicates the relationship between current assets (sources of repayment of liabilities) and current liabilities (financial commitments) .

$$\text{CURRENT RATIO} = \frac{\text{CURRENT ASSETS}}{\text{CURRENT LIABILITIES}}$$

Data of Current Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	2.48	2.36	2.20
WIPRO	22.21	34.52	35.78
INFOSYS	1.88	1.89	2.62

SOURCE - <https://www.moneycontrol.com/india/screner.com>



CURRENT RATIO OF TCS

(as on Mar 2023)

Current Assets – 94,918 Cr

Current Liabilities – 43,061Cr

$$\begin{aligned}\text{CURRENT RATIO} &= \frac{\text{CURRENT ASSETS}}{\text{CURRENT LIABILITIES}} \\ &= \frac{94,918}{43,061} \\ &= 2.20\end{aligned}$$

CURRENT RATIO OF WIPRO

(as on Mar 2023)

Current Assets – 661,096 Cr

Current Liabilities – 18,445.40 Cr

$$\begin{aligned}\text{CURRENT RATIO} &= \frac{\text{CURRENT ASSETS}}{\text{CURRENT LIABILITIES}} \\ &= \frac{661,096}{18,445.40} \\ &= 35.78\end{aligned}$$

CURRENT RATIO OF WIPRO

(as on Mar 2024)

Current Assets – 70,952Cr

Liabilities – 27,086 Cr

$$\begin{aligned}\text{CURRENT RATIO} &= \frac{\text{CURRENT ASSETS}}{\text{CURRENT LIABILITIES}} \\ &= \frac{70,952}{27,086} \\ &= 2.62\end{aligned}$$

Comparison of current Ratio

COMPANIES	CURRENT ASSETS	CURRENT LIABILITIES	CURRENT RATIO
TCS	94,918	2,84,516	2.52
WIPRO	661,096	18,445.40	35.78
INFOSYS	70,952	27,086	2.62

INTERPRETATION

- **Trend:** Wipro's current ratio has generally been above 1 throughout the period, indicating that the company has sufficient current assets to cover its short-term obligations.
- **Comparison:** A current ratio of around 2 is generally considered healthy. Wipro's current ratio has fluctuated slightly, but it has consistently remained above this benchmark.
- **Factors Affecting Current Ratio:** Factors such as inventory levels, accounts receivable, and accounts payable can significantly impact the current ratio.

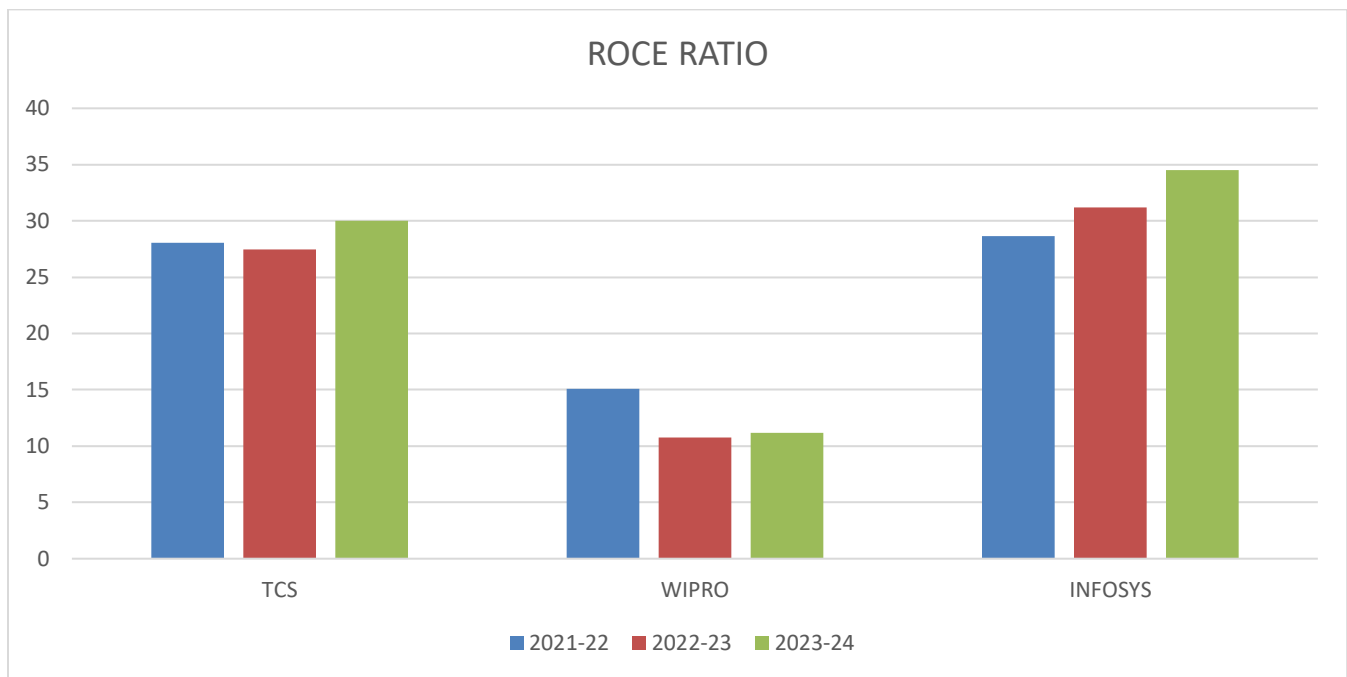
5. **RETURN ON CAPITAL EMPLOYED** : Return on Capital Employed (ROCE) is a financial metric that measures the profitability of a company in relation to its overall capital, which includes both equity AND equity

$$\text{RETURN ON CAPITAL EMPLOYED} = \frac{\text{NET INCOME}}{\text{AVERAGE TOTAL ASSETS}}$$

Data of ROCE for last 3 years

	2021-22	2022-23	2023-24
TCS	28.06	27.45	30.03
WIPRO	15.09	10.75	11.16
INFOSYS	28.67	31.21	34.49

SOURCE - <https://www.moneycontrol.com/india/www.screener.com>



CURRENT ROCE OF TCS

EBIT - 3,15,752

Capital Employed –1,01,989

$$\text{ROCE} = \frac{\text{EBIT}}{\text{Capital Employed}}$$

$$\begin{aligned}
 & \frac{58,560}{1,01,989} \\
 = & 30.03\%
 \end{aligned}$$

CURRENT ROCE OF WIPRO

EBIT - 12,417.50

Capital Employed – 81,648.60

$$\begin{aligned}
 \text{ROCE} &= \frac{\text{EBIT}}{\text{Capital Employed}} \\
 &= \frac{12,417.50}{81,648.60} \\
 &= 11.16
 \end{aligned}$$

CURRENT ROCE OF INFOSYS

EBIT - 35,953

Capital Employed – 81,648.60

Capital Employed = Total Assets - Current Liabilities
 = 114,950 - 27,086
 = 87,864

$$\begin{aligned}
 \text{ROCE} &= \frac{\text{EBIT}}{\text{Capital Employed}} \\
 &= \frac{35,953}{87,864}
 \end{aligned}$$

= 34.49

Comparison of ROCE

Companies	EBIT Cr	Capital Employed	ROCE (%)
TCS	58,560	1,01,989	30.03
WIPRO	12,648.60	81,648.60	11.16
INFOSYS	35,953	87,864	34.49

INTERPRETATION

ROCE: A financial ratio that measures how efficiently a company uses its capital to generate profits.

- **TCS:** With an ROCE of 30.03%, TCS has demonstrated superior efficiency in utilizing its capital to generate profits. This suggests that TCS is effectively managing its investments and maximizing returns.
- **Wipro:** While Wipro's ROCE of 11.16% is respectable, it is significantly lower than TCS's. This indicates that Wipro may be less efficient in capital allocation or facing challenges in generating returns from its investments.
- **Infosys:** With an ROCE of 34.49%, Infosys has demonstrated superior efficiency in utilizing its capital to generate profits. This suggests that Infosys is more effectively managing its investments and maximizing returns.
-

Factors Affecting ROCE:

- **Operating Efficiency:** Companies with higher operating efficiency tend to have higher ROCEs.
- **Asset Turnover:** A high asset turnover ratio (sales/total assets) can contribute to a higher ROCE.
- **Profit Margins:** Higher profit margins can also boost ROCE.

6. **RETURN ON EQUITY** : It measures the amount of profit that the company generates on the money invested by the shareholders. It also helps you understand how a company compares to other firms in the same industry and evaluate the company's financial performance and asset valuation.

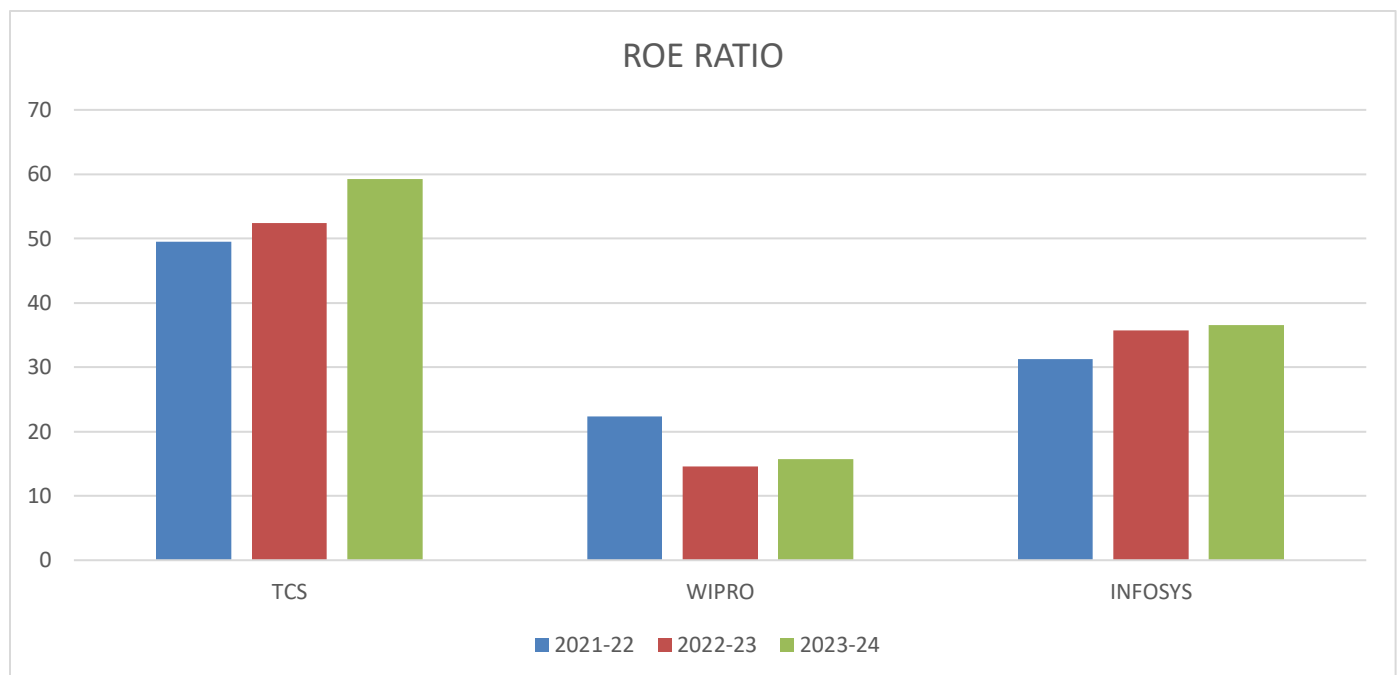
$$\text{RETURN ON EQUITY} = \frac{\text{PROFIT AFTER TAX}}{\text{NET INCOME}} \times 100$$

Data of ROE for last 3 years

	2021-22	2022-23	2023-24
TCS	49.47	52.44	59.22
WIPRO	22.31	14.61	15.76
INFOSYS	31.24	35.75	36.55

SOURCE-

<https://www.moneycontrol.com/india/>
www.screener.c



CURRENT ROE OF TCS

Net Income (in crores) 43,559

Average Shareholders' Equity (in crores)- 73,644

$$\begin{aligned}\text{ROE} &= \frac{\text{Net Income}}{\text{Average Shareholders' Equity}} \\ &= \frac{43,559}{73,644} \\ &= 59.22 \%\end{aligned}$$

CURRENT ROE OF WIPRO

NET INCOME Cr - 9,118.60

AVERAGE SHAREHOLDER'S
EQUITY Cr – 57,781.90

$$\begin{aligned}\text{ROE} &= \frac{\text{Net Income}}{\text{Average Shareholders' Equity}} \\ &= \frac{9,11.60}{57,781.90} \\ &= 15.76 \%\end{aligned}$$

CURRENT ROE OF INFOSYS

NET INCOME Cr - 9,118.60

AVERAGE SHAREHOLDER'S
EQUITY Cr – 57,781.90

$$\text{ROE} = \frac{\text{Net Income}}{\text{Average Shareholders' Equity}}$$

$$= \frac{27,234}{74,458.5}$$

$$= 36.55 \%$$

COMPARISON OF ROE

Companies	Net Income (in crores)	Average Shareholders' Equity (in crores)	ROE (%)
TCS	43,559	73,644	59.22
WIPRO	9,118.60	57,781.90	15.76
INFOSYS	27,234	74,458.5	36.55

INTERPRETATION

ROE: A financial ratio that measures how efficiently a company uses its shareholders' equity to generate profits.

- **TCS:** With an ROE of 36.55 %, TCS has demonstrated superior efficiency in utilizing its shareholders' equity to generate profits. This suggests that TCS is effectively managing its capital and maximizing returns for its investors.
- **Wipro:** While Wipro's ROE of 15.76% is respectable, it is significantly lower than TCS's. This

indicates that Wipro may be less efficient in capital allocation or facing challenges in generating returns from its equity investments.

- **Infosys:** With an ROE of 59.22%, infosys has demonstrated superior efficiency in utilizing its shareholders' equity to generate profits. This suggests that TCS & Infosys is effectively managing its capital and maximizing returns for its investors

Factors Affecting ROE:

- **Profitability:** Higher net income relative to equity leads to a higher ROE.
- **Asset Turnover:** Efficient use of assets can boost ROE.
- **Financial Leverage:** A higher debt-to-equity ratio can amplify returns (both positive and negative).

7. **NET PROFIT MARGIN** : Net Profit Margin is a direct measure of a company's profitability. It represents the percentage of revenue that translates into net profit after all expenses, including taxes and interest, have been accounted for.

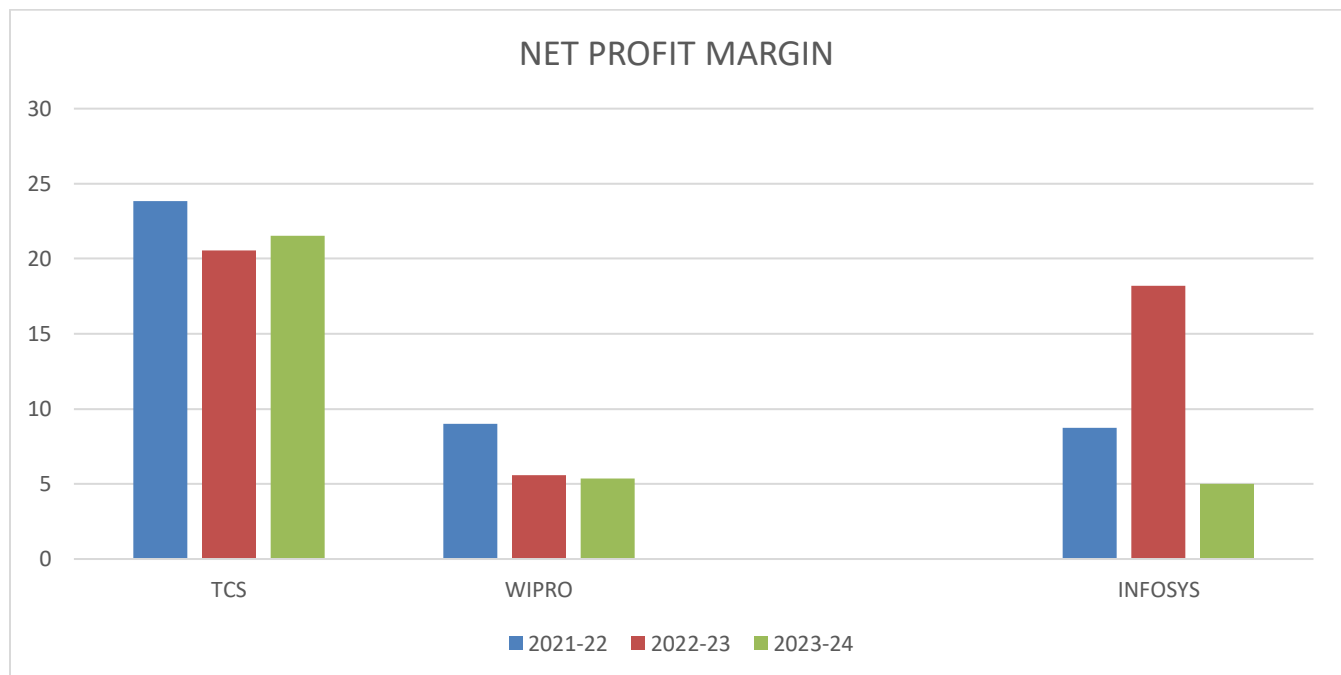
$$\text{NET PROFIT MARGIN} = \frac{\text{NET PROFIT}}{\text{NET SALES}}$$

Data of Net Profit Margin for last 3 years

	2021-22	2022-23	2023-24
TCS	23.82	20.54	21.53
WIPRO	8.99	5.59	5.36
INFOSYS	8.72	18.21	19.95

SOURCE-

<https://www.moneycontrol.com/india/screner.c>



NET PROFIT MARGIN OF TCS

NET PROFIT - 43,559

NET SALES - 202,359

$$\text{Net Profit Margin} = \frac{\text{NET PROFIT}}{\text{NET SALES}}$$

$$= \frac{43,559}{202,359}$$

$$= 21.53$$

NET PROFIT MARGIN OF WIPRO

NET PROFIT - 9,118.60

NET SALES - 170,155

$$\text{Net Profit Margin} = \frac{\text{NET PROFIT}}{\text{NET SALES}}$$

$$= \frac{9,11.60}{170,155}$$

$$= 5.36 \%$$

NET PROFIT MARGIN OF INFOSYS

NET PROFIT - 9,118.60

NET SALES - 170,155

$$\text{Net Profit Margin} = \frac{\text{NET PROFIT}}{\text{NET SALES}}$$

$$= \frac{27,234}{136,350}$$
$$= 19.95 \%$$

Comparison of NET PROFIT MARGIN Ratio

COMPANIES	Net Sales (in crores)	Net Profit (in crores)	Net Profit Margin (%)
TCS	202,359	43,559	21.53
WIPRO	9,11.60	170,155	5.36
INFOSYS	136,350	27,234	19.95

INTERPRETATION

Net Profit Margin: A financial ratio that measures the percentage of net profit a company generates from its revenue.

- **TCS:** With a net profit margin of 21.53%, TCS has a significantly higher profitability compared to Wipro. This suggests that TCS is more efficient in managing its costs and generating profits from its operations.
- **Wipro:** Wipro's net profit margin of 5.36% is relatively lower, indicating that the company may face

challenges in controlling costs or increasing revenue.

- **Infosys :** With a net profit margin of 19.95 %, TCS has a significantly higher profitability compared to Wipro. This suggests that TCS is more efficient in managing its costs and generating profits from its operations.

-

Factors Affecting Net Profit Margin:

- **Operating Expenses:** Higher operating expenses can reduce net profit margin.
- **Pricing Strategy:** Higher pricing can increase net profit margin, but it may also affect demand.
- **Tax Rates:** Changes in tax rates can impact net profit margin.

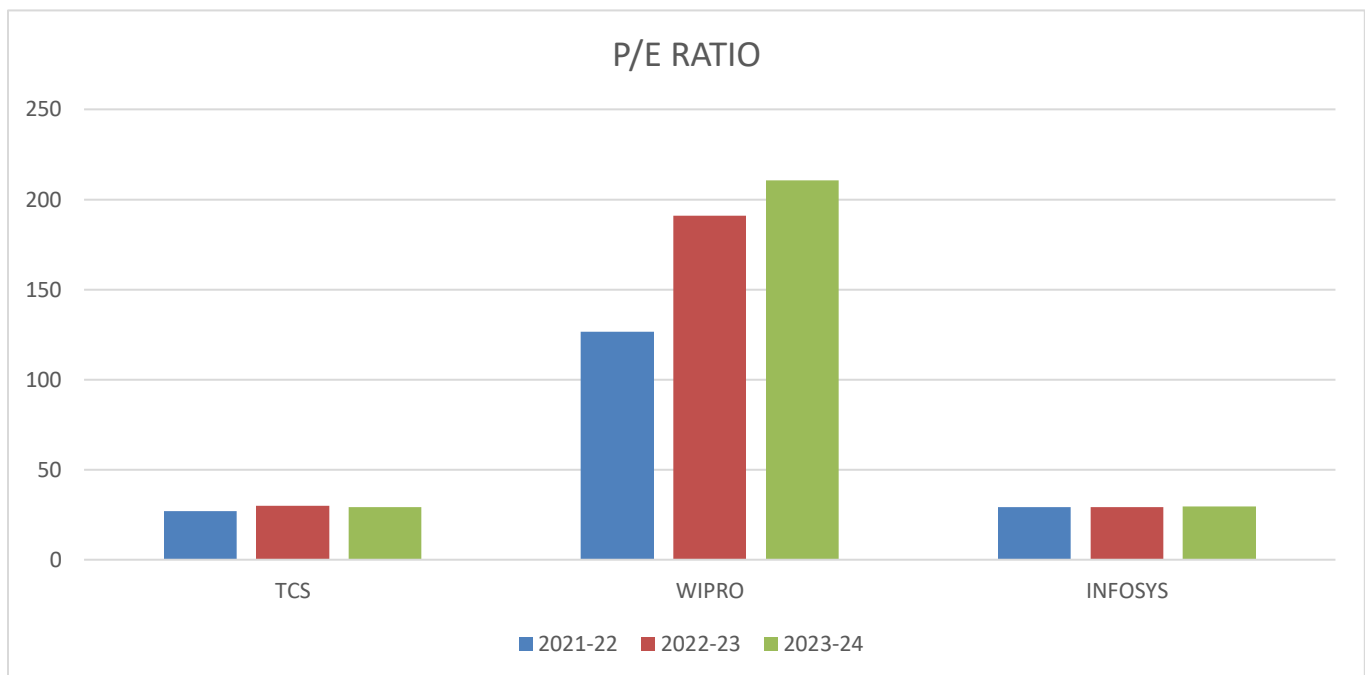
8. **PRICE TO EARNING RATIO** - Price to Earnings Ratio (P/E Ratio) is the ratio of share price of a stock to its earnings per share (EPS). PE ratio is one of the most popular valuation metrics of stocks. It provides indication whether a stock at its current market price is expensive or cheap.

$$\text{P/E RATIO} = \frac{\text{MARKET VALUE PER SHARE}}{\text{TTM EPS}}$$

Data of P/E Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	27.08	29.86	29.33
WIPRO	126.45	190.91	210.54
INFOSYS	29.03	29.32	29.47

SOURCE - <https://www.moneycontrol.com/india/>
www.screener.com



CURRENT P/E RATIO OF TCS

Market price (as on -27-09-2024) – 35,00

EPS TTM (as on -27-09-2024) – 54.82

$$\text{P/E Ratio} = \frac{\text{Market value per share}}{\text{TTM EPS}}$$

$$\begin{aligned} &= \frac{35,00}{54.82} \\ &= 63.86 \end{aligned}$$

CURRENT P/E RATIO OF WIPRO

Mkt price (as on -27-09-2024) –542

EPS TTM (as on 27-09-2024) – 16.63

$$\text{P/E Ratio} = \frac{\text{Market value per share}}{\text{TTM EPS}}$$

$$\begin{aligned} &= \frac{542}{35.97} \\ &= 15.07 \end{aligned}$$

CURRENT P/E RATIO OF INFOSYS

Mkt price (2024) – 1934

EPS TTM (2024) – 65.62

$$\begin{aligned}\text{P/E Ratio} &= \frac{\text{Market value per share}}{\text{TTM EPS}} \\ &= \frac{19,34}{65.62} \\ &= 29.47\end{aligned}$$

Comparison of P/E Ratio

COMPANIES	MARKET VALUE PER SHARE	TTM EPS	P/E RATIO
TCS	35,00	54.86	63.86
WIPRO	542	35.97	15.07
INFOSYS	19,34	65.62	29.47

Valuation as on 27-09-2024

COMPANIES	CURRENT MARKET PRICE	P/E	VALUATION
TCS	3,500	63.86	Overvalued
WIPRO	542	15.07	Undervalued
INFOSYS	1934	29.47	Overvalued
IDEAL Industry P/E RATIO		20-25	

If Current P/E < Industry P/E = Undervalued

TCS – $63.86 > 25$ = Overvalued

WIPRO – $15.07 < 25$ = undervalued

INFOSYS - $29.47 < 25$ = Overvalued

INTERPRETATION

- **TCS:** The lower P/E ratio for WIPRO suggests that it may be relatively undervalued compared to TCS. However, it's important to consider other factors such as growth prospects and industry benchmarks.
- **Wipro:** The higher P/E ratio for TCS indicates that investors are willing to pay a premium for its future earnings growth. However, this also means that the stock is more susceptible to market fluctuations.
- **Infosys :** The lower P/E ratio for WIPRO suggests that it may be relatively undervalued compared to Infosys. However, it's important to consider other factors such as growth prospects and industry benchmarks.

- A higher P/E ratio generally suggests that investors are willing to pay a premium for a company's future earnings growth. In this case, TCS's P/E ratio of 63.86 is significantly higher than WIPRO's P/E ratio of 15.07. This suggests that investors are placing a much higher valuation on TCS's future earnings potential compared to WIPRO.

9. **DIVIDEND PAY OUT RATIO** = The Dividend Payout Ratio measures the proportion of a company's net income that it distributes to shareholders as dividends. A high payout ratio indicates that a company is returning a significant portion of its earnings to shareholders, while a low payout ratio suggests that the company is retaining a larger portion of its earnings for reinvestment.

DIVIDEND PAY OUT RATIO = DIVIDEND PAID / NET INCOME

DIVIDEND PAY OUT RATIO OF TCS

DIVIDEND PAID - 25,137

NET INCOME - 43,559

$$\begin{aligned}
 \text{DIVIDEND PAY OUT} &= \frac{\text{DIVIDEND PAID}}{\text{NET INCOME}} \\
 &= \frac{25,137}{43,559} \\
 &= 57.70 \%
 \end{aligned}$$

PAY OUT RATIO OF WIPRO

DIVIDEND PAID - 25,137

NET INCOME - 43,559

$$\begin{aligned}\text{DIVIDEND PAY OUT} &= \frac{\text{DIVIDEND PAID}}{\text{NET INCOME}} \\ &= \frac{25,137}{43,559} \\ &= 0.577\end{aligned}$$

PAY OUT RATIO OF INFOSYS

DIVIDEND PAID - 14.73

NET INCOME - 65.62

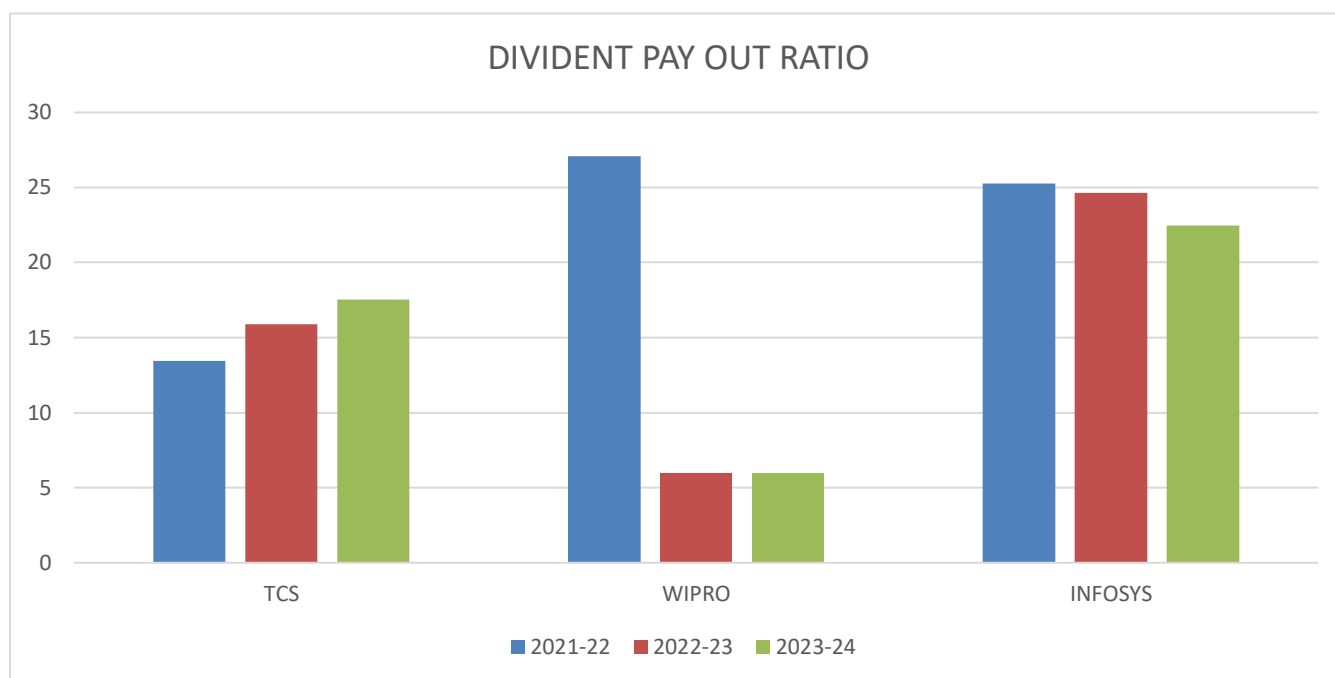
$$\begin{aligned}\text{DIVIDEND PAY OUT} &= \frac{\text{DIVIDEND PAID}}{\text{NET INCOME}} \\ &= \frac{14.73}{65.62} \\ &= 0.224\end{aligned}$$

Data of DIVIDEND PAY OUT RATIO Ratio for last 3 years

Data of Dividend payout for last 3 years

	2021-22	2022-23	2023-24

TCS	13.44	15.87	17.53
WIPRO	27.07	5.97	5.97
INFOSYS	25.25	24.65	22.47



Comparison of Dividend payout

COMPANIES	Dividends Paid (in crores)	Net Income (in crores)	Dividend Payout Ratio (%)
TCS	25,137	43,559	57.70

WIPRO	545.50	9,118.60	5.9
INFOSYS	14.73	65.62	22.47

INTERPRETATION

Dividend Payout Ratio (DPR): A financial metric that measures the percentage of net income a company distributes to shareholders as dividends.

- **TCS:** With a DPR of 57.70%, TCS distributes a significant portion of its net income to shareholders as dividends. This indicates a shareholder-friendly approach. However, a high payout ratio can also limit a company's ability to reinvest in growth.
- **Wipro:** With a DPR of 5.97%, Wipro retains a larger portion of its earnings for reinvestment. This may indicate a focus on growth and expansion, but it could also limit dividend income for shareholders.
- **INFOSYS:** With a DPR of 22.47%, Infosys distributes a significant portion of its net income to shareholders as dividends. This indicates a shareholder-friendly approach. However, a high payout ratio can also limit a company's ability to reinvest in growth.
-
- **Factors Affecting DPR:**
- **Earnings Growth:** Companies with strong earnings growth may choose to retain more profits for reinvestment.
- **Financial Health:** Companies with a strong financial position may have more flexibility to pay dividends.
- **Industry Practices:** Dividend payout policies can vary by industry

10. PRICE TO BOOK RATIO : The price-to-book ratio (P/B ratio) is a financial ratio used to compare a company's current market value to its book value (where book value is the value of all assets minus liabilities owned by a company).

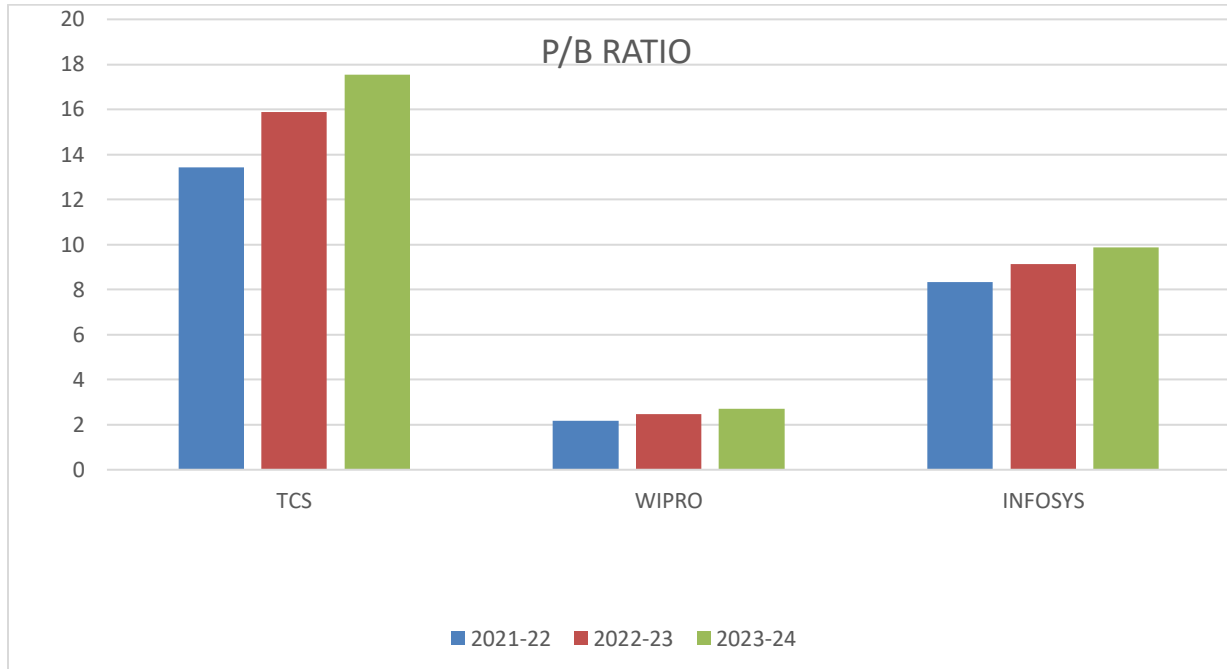
$$\text{P/B RATIO} = \frac{\text{MARKET PRICE PER SHARE}}{\text{BOOK VALUE PER SHARE}}$$

$$\text{Book Value} = \frac{\text{Total Assets} - \text{Total Liabilities}}{\text{Total Outstanding Shares}}$$

Data of P/B Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	13.44	15.87	17.53
WIPRO	2.16	2.48	2.71
INFOSYS	8.32	9.12	9.86

SOURCE - <https://www.moneycontrol.com/india/www.screener.com>



CURRENT P/B RATIO OF

TCS

Mkt – 3,500

B/V – Rs. 199.75

$$\begin{aligned}\text{P/B Ratio} &= \frac{\text{Market value per share}}{\text{Book value per share}} \\ &= \frac{3,500}{199.75} \\ &= 17.53\end{aligned}$$

CURRENT P/B OF WIPRO

_Mkt price – 542

B/V – Rs. 199.75

$$\begin{aligned}\text{P/B ratio} &= \frac{\text{Market value per share}}{\text{Book value per share}} \\ &= \frac{542}{199.75} \\ &= 2.71\end{aligned}$$

CURRENT P/B OF INFOSYS

_Mkt price – 1934

B/V – Rs. 199.75

$$\begin{aligned}\text{P/B ratio} &= \frac{\text{Market value per share}}{\text{Book value per share}}\end{aligned}$$

$$= 19,34$$

$$199.75$$

$$= 9.68$$

Valuation as on 27-09-2024

COMPANIES	CURRENT MARKET PRICE	P/B	VALUATION
TCS	3,500	17.53	Overvalued
WIPRO	542	2.71	Undervalued
INFOSYS	1934	9.68	Overvalued

If $P/B > MP$ = Overvalued

If $P/B < MP$ = Undervalued

If $P/B > MP$ = Overvalued

INTERPRETATION

- **Trend:** Both TCS, Wipro & INFOSYS have seen fluctuations in their P/B ratios over the past 3 years, reflecting changes in market sentiment and expectations for their future earnings.
- **Comparison:** TCS generally has a higher P/B ratio compared to Wipro & INFOSYS, suggesting that investors are willing to pay a premium for TCS's assets.

Factors Affecting P/B Ratio: Factors such as growth prospects, asset quality, and market sentiment can influence the P/B ratio.

- Both TCS and Wipro have experienced changes in their P/B ratios over the past five years.
- TCS consistently has a higher P/B ratio than Wipro, indicating a higher market valuation.
- It's important to consider other valuation metrics and factors when analysing the stocks of TCS and Wipro.

11. PRICE TO EARNING GROWTH – The Price to earnings growth (PEG Ratio) assesses a stock's valuation by considering both its P/E ratio and earnings growth. APEG ratio below 1 generally suggests that the stock may be undervalued, indicating that the market is not fully pricing in the company's expected earnings growth.

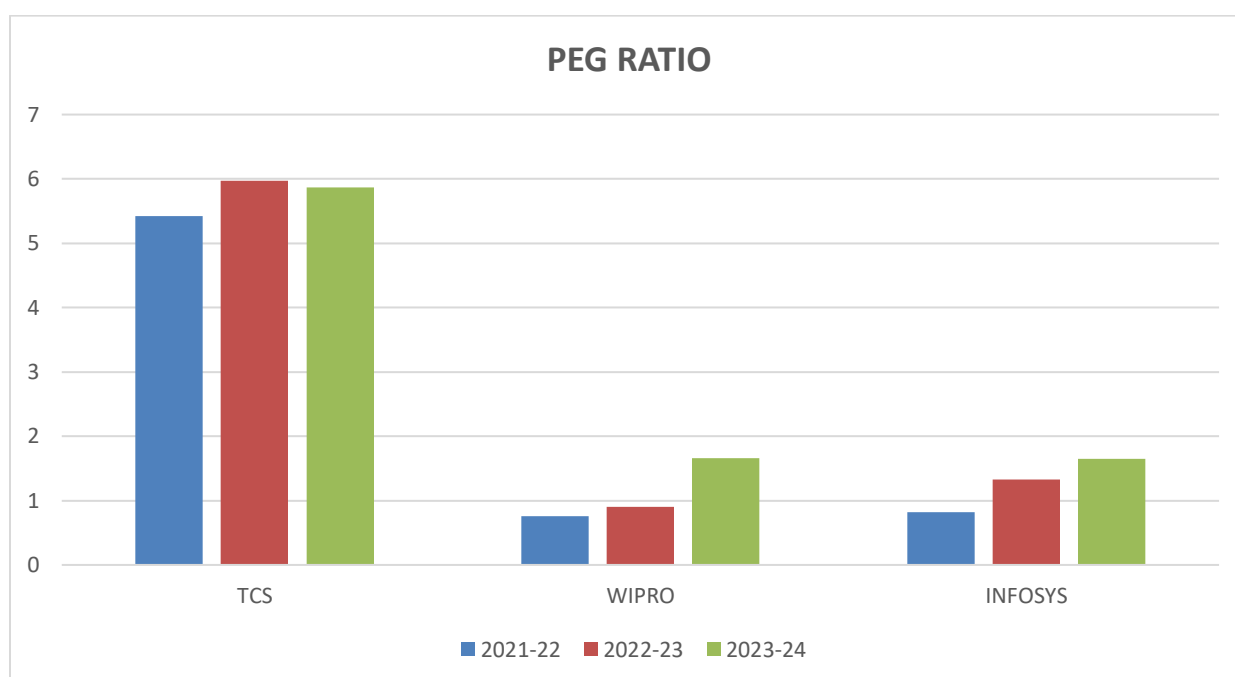
$$\text{PEG RATIO} = \text{P/E RATIO} / \text{ESTIMATED EPS GROWTH RATE}$$

Estimated EPS Growth Rate = Average Growth Rate of EPS P/E Ratio = TTM P/E Ratio

Data of PEG Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	5.42	5.97	5.87
WIPRO	0.76	0.90	1.66
INFOSYS	0.82	1.33	1.65

SOURCE - <https://www.moneycontrol.com/india/wwww.screener.com>



CURRENT PEG RATIO OF TCS

P/E Ratio – 24.9

$$\begin{aligned}\text{Estimated EPS Growth rate} &= \frac{\text{Current Year EPS} - \text{Last Year EPS}}{\text{Current Year EPS}} \\ &= \frac{54.82 - 71.25}{54.82} \\ &= \frac{16.43}{54.82} \\ &= 0.2910\end{aligned}$$

$$\begin{aligned}\text{PEG Ratio} &= \frac{\text{P/E Ratio}}{\text{Estimated EPS Growth rate}} \\ &= \frac{0.2910}{15} \\ &= 0.01910\end{aligned}$$

CURRENT PEG RATIO OF WIPRO

P/E Ratio – 19.9

$$\text{Estimated EPS Growth rate} = \frac{\text{Current Year EPS} - \text{Last Year EPS}}{\text{Current Year EPS}}$$

$$\begin{aligned}
 &= \frac{16.63 - 16.75}{16.63} \\
 &= \frac{-0.12}{16.63} \\
 &= 0.00722
 \end{aligned}$$

$$\begin{aligned}
 \text{PEG Ratio} &= \frac{\text{P/E Ratio}}{\text{Estimated EPS Growth rate}} \\
 &= \frac{-0.00722}{15/9.12} \\
 &= 1.66
 \end{aligned}$$

CURRENT PEG RATIO OF WIPRO

P/E Ratio – 29.47

$$\begin{aligned}
 \text{PEG Ratio} &= \frac{\text{P/E Ratio}}{\text{Estimated EPS Growth rate}} \\
 &= \frac{29.65}{17.85} \\
 &= 1.65
 \end{aligned}$$

Valuation as on 27.09-2024

Companies	P/E Ratio	Earnings Growth Rate (%)	PEG Ratio
TCS	6.36	0.23	27.65
WIPRO	24.9	159.12	1.66
INFOSYS	29.47	17.85	1.65

INTERPRETATION

- **Trend:** Wipro's PEG ratio has fluctuated over the past five years, with a significant decline in Mar-21 and an increase in subsequent years.
- **Comparison:** A PEG ratio below 1 is generally considered favourable, suggesting that the stock is undervalued relative to its growth prospects.
- **Factors Affecting PEG Ratio:** Factors such as market sentiment, economic conditions, and the company's specific growth prospects can influence the PEG ratio.

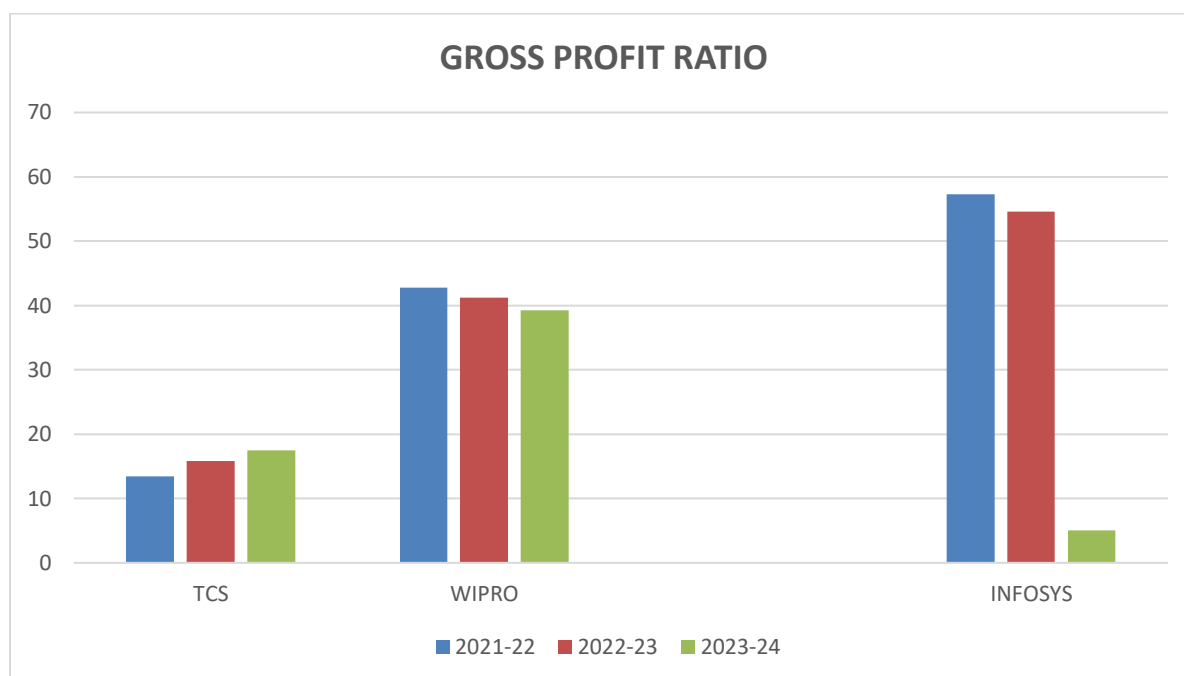
12. **GROSS PROFIT RATIO** : The Gross Profit Ratio is a financial metric that measures a company's profitability after deducting the cost of goods sold (COGS) from its revenue. It indicates the company's efficiency in managing its direct costs associated with producing or selling its products or services.

GROSS PROFIT RATIO = GROSS PROFIT / NET SALES

Data of Gross profit Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	13.44	15.87	17.53
WIPRO	42.76	41.26	39.25
INFOSYS	57.31	54.63	51.75

SOURCE - <https://www.moneycontrol.com/india/www.screener.com>



GROSS PROFIT RATIO OF TCS

P/E Ratio – 29.33

$$\begin{aligned}\text{Estimated EPS Growth rate} &= \frac{\text{COST OF GOODS SOLD} - \text{GROSS PROFIT}}{\text{NET SALES}} \\ &= \frac{151,072 - 51,287}{202,359} \\ &= \frac{99,785}{202,359} \\ &= 0.4931\end{aligned}$$

GROSS PROFIT RATIO OF WIPRO

P/E Ratio – 29.33

$$\begin{aligned}\text{Estimated EPS Growth rate} &= \frac{\text{COST OF GOODS SOLD} - \text{GROSS PROFIT}}{\text{NET SALES}} \\ &= \frac{66,792.40}{170,155} \\ &= \frac{39.25}{100}\end{aligned}$$

GROSS PROFIT RATIO OF INFOSYS

P/E Ratio – 29.33

$$\begin{aligned}\text{Estimated EPS Growth rate} &= \frac{\text{COST OF GOODS SOLD} - \text{GROSS PROFIT}}{\text{NET SALES}} \\ &= \frac{35,953}{128,933} \\ &= \frac{29.90}{100}\end{aligned}$$

COMPARISON OF Gross Profit ratio

	Net Sales (in crores)	Cost of Goods Sold (in crores)	Gross Profit (in crores)	Gross Profit Ratio (%)
TCS	202,359	151,072	51,287	25.33
WIPRO	170,155	66,792.40	34,346	39.25
INFOSYS	35,953	124,014	128,933	27.90

INTERPRETATION

- **TCS:** With a Gross Profit Ratio of 25.33%, TCS has a lower gross profit margin compared to Wipro. This could indicate that TCS faces higher costs of goods sold or has a less profitable product mix.
- **Wipro:** Wipro's Gross Profit Ratio of 39.25% suggests that it is more efficient in managing its COGS and generating profits from its core operations.
- **Infosys :** With a Gross Profit Ratio of 27.90 %, infosys has a lower gross profit margin compared to Wipro. This could indicate that INFOSYS faces higher costs of goods sold or has a less profitable product mix.
-

Factors Affecting Gross Profit Ratio:

- **Pricing Strategy:** Higher pricing can increase Gross Profit Ratio.
- **Cost Management:** Efficient management of COGS can improve Gross Profit Ratio.
- **Product Mix:** A shift towards more profitable products can increase Gross Profit Ratio.

13.Working Capital Ratio: measures a company's ability to pay short-term debts using its

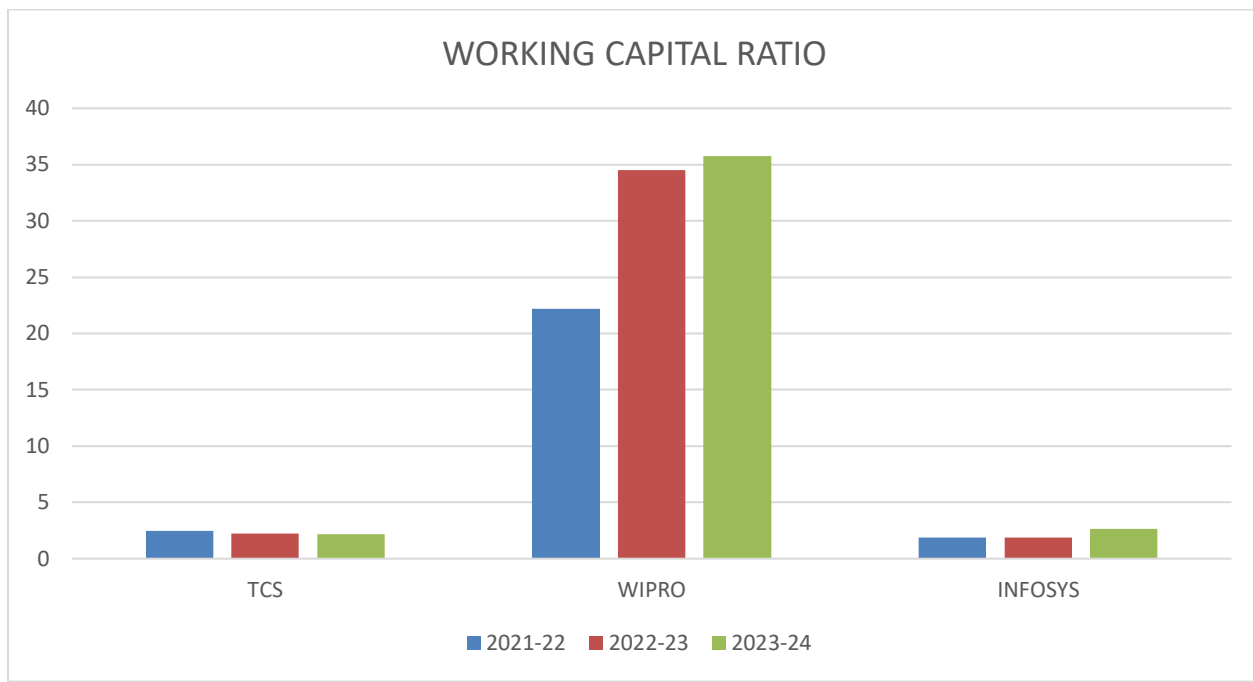
current assets. A ratio greater than 1 is generally good, indicating sufficient liquidity. The working capital ratio is a financial ratio that measures a company's ability to pay its short-term liabilities using its current assets. It is calculated by dividing a company's current assets by its current liabilities.

$$\text{Working Capital Ratio} = \text{Current Assets} / \text{Current Liabilities}$$

Data of Working Capital Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	2.48	2.26	2.20
WIPRO	22.21	34.52	35.78
INFOSYS	1.88	1.89	2.62

SOURCE - <https://www.moneycontrol.com/india/screner.com>



WORKING CAPITAL RATIO OF TCS

CURRENT ASSET = 94,918

CURRENT LIABILITIES = 43,061

$$\begin{aligned}
 \text{WORKING CAPITAL} &= \frac{\text{CURRENT ASSET}}{\text{CURRENT LIABILITIES}} \\
 &= \frac{94,918}{43,061} \\
 &= 2.20
 \end{aligned}$$

WORKING CAPITAL RATIO OF WIPRO

CURRENT ASSET = 661,096

CURRENT LIABILITIES = 18,445.40

$$\begin{aligned}
 \text{WORKING CAPITAL} &= \frac{\text{CURRENT ASSET}}{\text{CURRENT LIABILITIES}} \\
 &= \frac{661.096}{18,445.40} \\
 &= 35.78
 \end{aligned}$$

WORKING CAPITAL RATIO OF INFOSYS

CURRENT ASSET = 70,952

CURRENT LIABILITIES = 27,086

$$\begin{aligned}
 \text{WORKING CAPITAL} &= \frac{\text{CURRENT ASSET}}{\text{CURRENT LIABILITIES}} \\
 &= \frac{70,952}{27,086} \\
 &= 2.26
 \end{aligned}$$

COMPARISON OF WORKING CAPITAL

COMPANIES	CURRENT ASSETS Cr	CURRENT LIABILITIES Cr	WORKING CAPITAL
TCS	94,918	43,061	2.20
WIPRO	661,096	18,445.40	35.78
INFOSYS	70,952	27,086	2.62

INTERPRETATION

- **TCS:** With a WCR of 2.20, TCS has a strong liquidity position, but it's slightly lower than Wipro's. This suggests that TCS might have a slightly tighter cash flow management.
- **Wipro:** Wipro's WCR of 35.78 indicates an even stronger liquidity position, suggesting that it has ample resources to cover its short-term obligations.
- **Infosys :** With a WCR of 2.62, Infosys has a strong liquidity position, but it's slightly lower than Wipro & TCS's. This suggests that Infosys might have a slightly tighter cash flow management.
-

Factors Affecting WCR:

- **Inventory Management:** Higher inventory levels can increase current assets, leading to a higher WCR.
- **Accounts Receivable:** A large volume of outstanding receivables can also increase WCR.
- **Accounts Payable:** A higher level of accounts payable can decrease WCR.

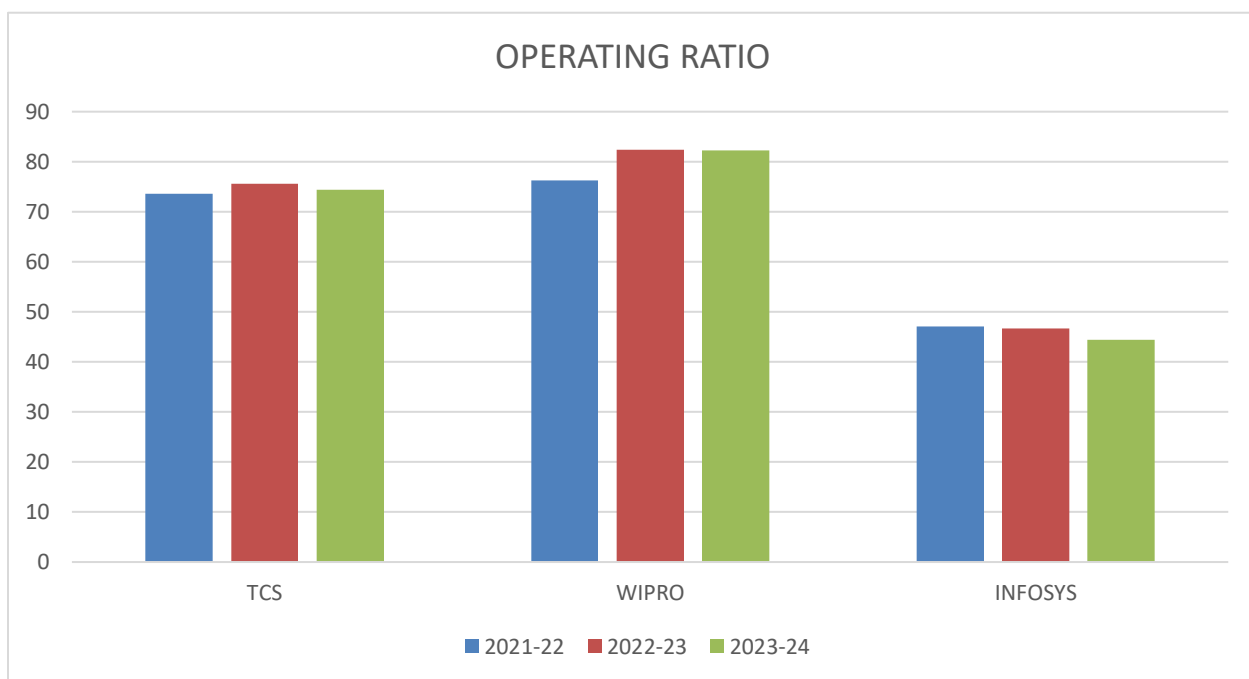
14.OPERATING RATIO : Operating Ratio measures how efficiently a company manages its day-to-day operations. A lower ratio means the company spends less to generate each dollar of revenue, which is generally a good sign.,_Operating Expenses: Include all costs related to a company's day-to-day operations, such as salaries, rent, utilities, and marketing expenses Net Sales: Total revenue from sales, minus any discounts, returns, or allowance

$$\text{OPERATING RATIO} = \text{OPERATING EXPENSES} / \text{NET SALES}$$

Data of OPERATION Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	73.63	75.65	74.36
WIPRO	76.27	82.45	82.23
INFOSYS	47.13	46.62	44.46

SOURCE - <https://www.moneycontrol.com/india/www.screener.com>



OPERATING RATIO OF TCS

OPERATING EXPENSES = 94,918

NET SALES = 43,061

$$\begin{aligned}\text{OPERATING RATIO} &= \frac{\text{OPERATING EXPENSES}}{\text{NET SALES}} \\ &= \frac{151,072}{202,359} \\ &= 74.36\end{aligned}$$

OPERATING RATIO OF WIPRO

OPERATING EXPENSES = 94,918

NET SALES = 43,061

$$\begin{aligned}\text{OPERATING RATIO} &= \frac{\text{OPERATING EXPENSES}}{\text{NET SALES}} \\ &= \frac{57,420.70}{69,838.20} \\ &= 82.23\end{aligned}$$

OPERATING RATIO OF INFOSYS

OPERATING EXPENSES = 58,420

NET SALES = 128,933

$$\begin{aligned}\text{OPERATING RATIO} &= \frac{\text{OPERATING EXPENSES}}{\text{NET SALES}} \\ &= \frac{58,420}{128,933}\end{aligned}$$

128,933

= 44.46

COMPARISON OF OPERATING RATIO

COMPANIES	Operating Expenses (in crores)	Net Sales (in crores)	Operating Ratio (%)
TCS	151,072	202,359	74.63
WIPRO	57,420.70	69,838.20	82.23
INFOSYS	58,420	128,933	44.46

INTERPRETATION

- **TCS:** With an operating ratio of 74.63%, TCS has a lower operating ratio compared to Wipro, indicating that it is more efficient in managing its operating expenses.
- **Wipro:** Wipro's operating ratio of 82.23% suggests that it spends a higher portion of its revenue on operating costs, which could impact its profitability.
- **Infosys :** With an operating ratio of 44.46 %, Infosys has a lower operating ratio compared to Wipro & TCS's, indicating that it is more efficient in managing its operating expenses

Factors Affecting Operating Ratio:

- **Cost Management:** Efficient management of operating expenses can lead to a lower operating ratio.
- **Pricing Strategy:** Higher pricing can reduce the operating ratio.
- **Product Mix:** A shift towards more profitable products can also improve the operating ratio.

15. QUICK PROFIT RATIO : The quick ratio is a financial ratio that measures a company's

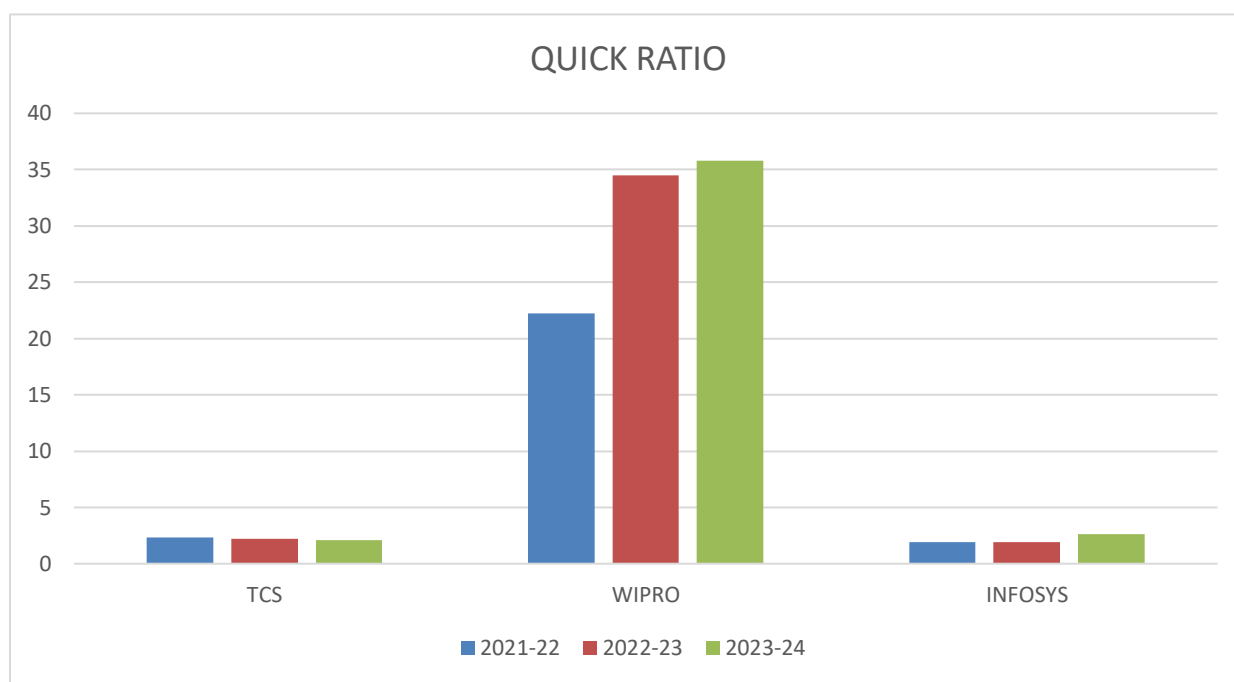
ability to pay its short-term liabilities using its most liquid assets. It is a more conservative measure of liquidity than the current ratio, as it excludes inventory from the calculation.

$$\text{Quick Ratio} = (\text{Cash} + \text{Marketable Securities} + \text{Accounts Receivable}) / \text{Current Liabilities}$$

Data of Working Quick Ratio for last 5 years

	2021-22	2022-23	2023-24
TCS	2.32	2.24	2.12
WIPRO	22.21	34.52	35.78
INFOSYS	1.89	1.90	2.63

SOURCE - <https://www.moneycontrol.com/india/www.screener.com>



QUICK RATIO OF TCS

Cash + Marketable Securities + Accounts Receivable = 91,466

CURRENT LIABILITIES = 43,061

$$\begin{aligned}\text{QUICK RATIO} &= \frac{\text{Cash + Marketable Securities + Accounts Receivable}}{\text{CURRENT LIABILITIES}} \\ &= \frac{91,466}{43,061} \\ &= 2.12\end{aligned}$$

QUICK RATIO OF WIPRO

$$\begin{aligned}\text{QUICK RATIO} &= \frac{\text{Cash + Marketable Securities + Accounts Receivable}}{\text{CURRENT LIABILITIES}} \\ &= \frac{661,023.1}{18,445.40} \\ &= 35.78\end{aligned}$$

QUICK RATIO OF INFOSYS

$$\begin{aligned}\text{QUICK RATIO} &= \frac{\text{Cash + Marketable Securities + Accounts Receivable}}{\text{CURRENT LIABILITIES}} \\ &= \frac{661,023.1}{18,445.40} \\ &= 35.78\end{aligned}$$

CURRENT LIABILITIES

$$= \frac{70,952}{27,086}$$

$$= 2.63$$

COMPARISON OF QUICK RATIO

COMPANIES	Cash + Marketable Securities + Accounts Receivable (in crores)	Current Liabilities (in crores)	QUICK Ratio (%)
TCS	91,466	43,061	2.12
WIPRO	661,023.1	18,445.40	35.78
INFOSYS	70,952	27,086	2.63

INTERPRETATION

- **TCS:** With a QR of 2.12, TCS has a strong liquidity position, but it's slightly lower than Wipro's. This suggests that TCS might have a slightly tighter cash flow management.
- **Wipro:** Wipro's QR of 35.78 indicates an even stronger liquidity position, suggesting that it has ample resources to cover its short-term obligations.
- **Infosys:** With a QR of 2.12, Infosys has a strong liquidity position, but it's slightly lower than Wipro & TCS. This suggests that Infosys might have a slightly tighter cash flow management.
-

Factors Affecting QR:

- **Inventory Management:** Higher inventory levels can decrease QR.

- **Accounts Receivable:** A large volume of outstanding receivables can increase QR.
- **Accounts Payable:** A higher level of accounts payable can decrease QR.

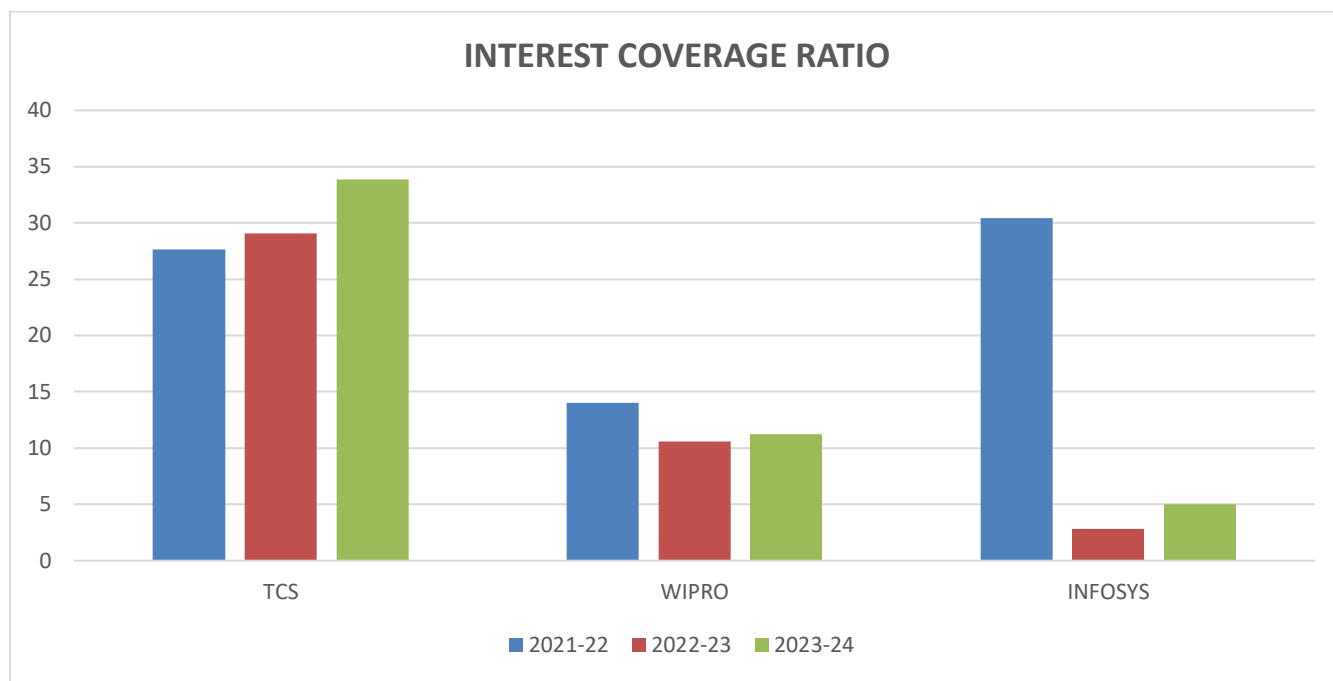
16. INTEREST COVERAGE RATIO : INTEREST COVERAGE RATIO MEANING WITH FORMULA IN SHORT ,A higher ICR indicates a company can easily cover its interest payments. A lower ICR suggests the company might struggle to meet its debt obligations. A generally accepted good ICR is 2 or higher, but it can vary by industry.

INTEREST COVERAGE RATIO = EBIT/INTEREST EXPENSES

Data of Interest Coverage Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	27.62	29.04	33.84
WIPRO	14.00	10.60	11.24
INFOSYS	30.48	31.32	32.41

SOURCE - <https://www.moneycontrol.com/india/wwww.screener.com>



INTEREST COVERAGE RATIO OF TCS

EBIT = 94,918

CURRENT LIABILITIES = 1,730

$$\begin{aligned}
 \text{INTEREST COVERAGE RATIO} &= \frac{\text{EBIT}}{\text{INTEREST EXPENSES}} \\
 &= \frac{58,560}{1,730} \\
 &= 33.84
 \end{aligned}$$

INTEREST COVERAGE RATIO OF WIPRO

EBIT = 12,417.50

INTEREST EXPENSES = 1,108.80

$$\text{INTEREST COVERAGE RATIO} = \frac{\text{EBIT}}{60}$$

INTEREST EXPENSES

$$\begin{aligned} &= \frac{12,417.50}{1,108.80} \\ &= 11.24 \\ &= 1 \end{aligned}$$

INTEREST COVERAGE RATIO OF INFOSYS

EBIT = 12,417.50

INTEREST EXPENSES = 1,108.80

$$\begin{aligned} \text{INTEREST COVERAGE RATIO} &= \frac{\text{EBIT}}{\text{INTEREST EXPENSES}} \\ &= \frac{1,109}{35,953} \\ &= 32.41 \end{aligned}$$

COMPARISON OF INTEREST COVERAGE RATIO

COMPANIES	EBIT (in crores)	Interest Expense (in crores)	Interest Coverage Ratio
TCS	58, 560	1,730	33.84
WIPRO	12,417.50	1,108.80	11.24

INFOSYS	35,953	1,109	32.41
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INTERPRETATION

- **TCS:** With an ICR of 33.84, TCS has a significantly stronger ability to cover its interest payments compared to Wipro. This indicates a lower risk of default on its debt obligations.
- **Wipro:** Wipro's ICR of 11.24 is also strong, but it's slightly lower than TCS. This suggests that TCS has a slightly greater financial cushion to handle potential interest rate increases or economic downturns.
- **Infosys :** With an ICR of 32.41, Infosys has a significantly stronger ability to cover its interest payments compared to Wipro. This indicates a lower risk of default on its debt obligations.

Factors Affecting ICR:

- **Interest Rates:** Rising interest rates can increase interest expenses, reducing the ICR.
- **Debt Levels:** A higher level of debt can increase interest expenses, lowering the ICR.
- **Operating Profitability:** Higher operating profits can increase the ICR.

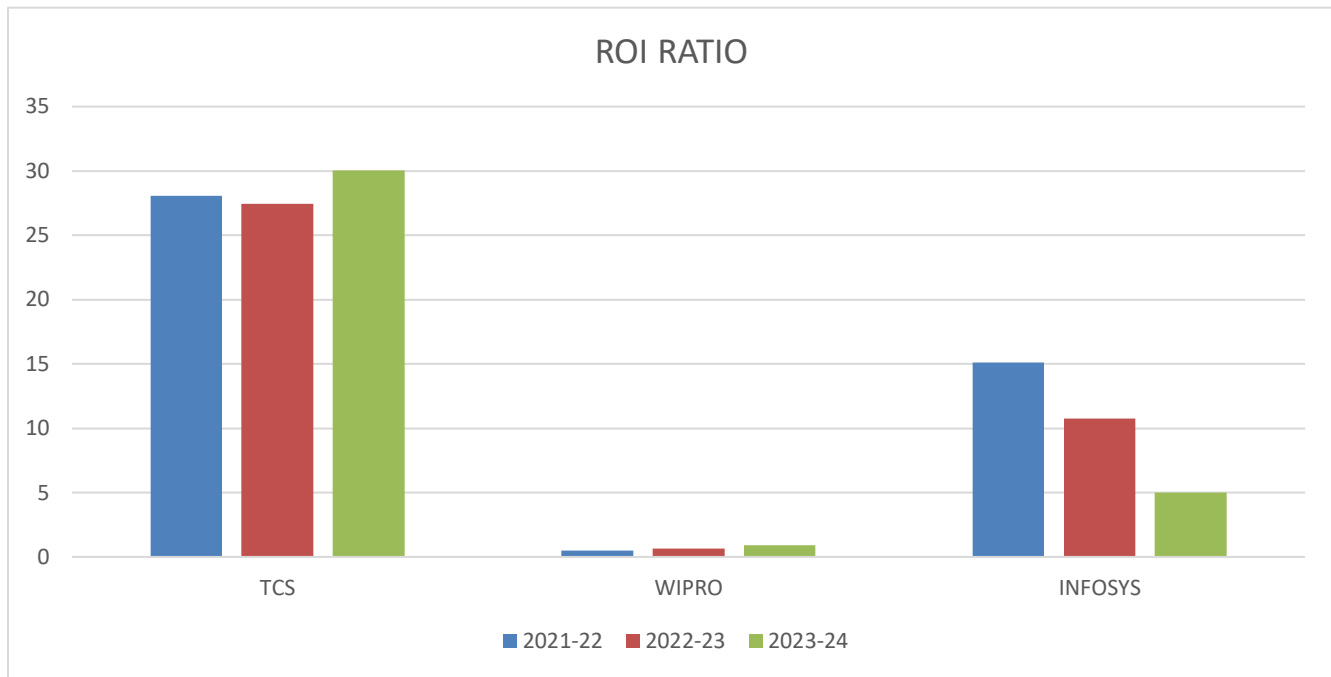
17.RETURN ON INVESTMENT RATIO : It's a financial metric that measures how much profit an investment generates relative to its cost. In simpler terms, it tells you how much you earned compared to how much you spent.

$$\text{RETURN ON INVESTMENT} = \text{NET PROFIT} / \text{INITIAL INVESTMENT}$$

Data of Return On Investment Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	28.06	27.45	30.03
WIPRO	0.50	0.67	0.89
INFOSYS	15.09	10.75	11.16

SOURCE - <https://www.moneycontrol.com/india/www.screener.com>



RETURN ON INVESTMENT RATIO OF TCS

PROFIT= 94,918

COST =

$$\begin{aligned}\text{RETURN ON INVESTMENT} &= \frac{\text{NET PROFIT}}{\text{INITIAL INVESTMENT}} \\ &= \frac{43,559}{145,050} \\ &= 30.03\end{aligned}$$

RETURN ON INVESTMENT RATIO OF WIPRO

PROFIT= 94,918

COST =

$$\begin{aligned}\text{RETURN ON INVESTMENT} &= \frac{\text{NET PROFIT}}{\text{INITIAL INVESTMENT}} \\ &= \frac{288,90}{9.71} \\ &= 23.57\end{aligned}$$

RETURN ON INVESTMENT RATIO OF INFOSYS

PROFIT=

COST =

$$\begin{aligned}\text{RETURN ON INVESTMENT} &= \frac{\text{NET PROFIT}}{\text{INITIAL INVESTMENT}} \\ &= \frac{27,234}{36.55} \\ &= 11.16 \%\end{aligned}$$

COMPARISON OF ROI

COMPANIES	Net Profit (in crores)	Average Total Assets (in crores)	ROI (%)
TCS	43,559	145,050	30.03
WIPRO	228,90	9.71	23.57
INFOSYS	27,234	36.55	11.16

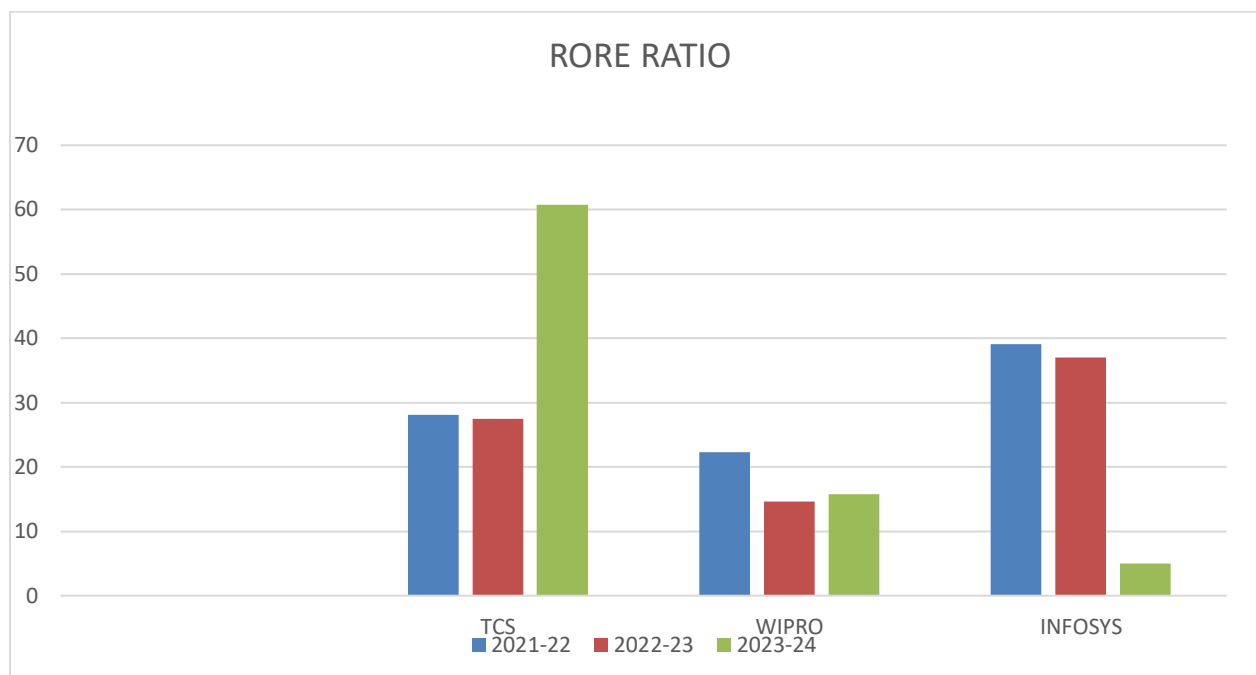
18.RETURN ON RETURNED EARNING RATIO: The Return on Returned Earnings (RORE) is a financial ratio that measures how efficiently a company reinvests its retained earnings to generate profits. It is calculated by dividing the net income of the current period by the accumulated retained earnings at the beginning of the period.

RETURN O N RETAINED EARNING RATIO = NET INCOME / RETAINED EARNING BRGINNING

Data of Return On Retained Ratio for last 3 years

	2021-22	2022-23	2023-24
TCS	28.06	27.45	60.73
WIPRO	22.31	14.61	15.76
INFOSYS	39.06	37.04	47.13

SOURCE - <https://www.moneycontrol.com/india/wwww.screener.com>



RETURN ON RETAINED EARNING RATIO OF TCS

NET INCOME= 43,559

RETURN ON RETAINED EARNING = 71,758

RETURN ON RETAINED = $\frac{\text{NET INCOME}}{\text{RETURN ON RETAINED EARNING}}$

EARNING RATION RETURN ON RETAINED EARNING

= $\frac{43,559}{71,758}$

71,758

= 60.73

RETURN ON RETAINED EARNING RATIO OF WIPRO

NET INCOME= 9,118.60

RETURN ON RETAINED EARNING = 57,781.90

$$\begin{aligned}
 \text{RETURN ON RETAINED EARNING RATION} &= \frac{\text{NET INCOME}}{\text{RETURN ON RETAINED EARNING}} \\
 &= \frac{9,118.60}{57,781.90} \\
 &= 15.76
 \end{aligned}$$

RETURN ON RETAINED EARNING RATIO OF INFOSYS

NET INCOME= 27,234

RETURN ON RETAINED EARNING = 57,781.90

$$\begin{aligned}
 \text{RETURN ON RETAINED EARNING RATION} &= \frac{\text{NET INCOME}}{\text{RETURN ON RETAINED EARNING}} \\
 &= \frac{27,234}{58,781.10} \\
 &= 47.13
 \end{aligned}$$

COMPARISON OF RORE

COMPANIES	Net Income (in crores)	Retained Earnings (Beginning (in crores))	RORE (%)
TCS	43,559	71,758	60.73
WIPRO	9,118.60	57,781.90	15.76
INFOSYS	27,234	58,781.10	47.13

INTERPRETATION

- **TCS:** With an RORE of 59.22%, TCS has demonstrated superior efficiency in utilizing its retained earnings to generate profits. This suggests that TCS is effectively reinvesting its earnings to drive growth and profitability.
- **Wipro:** Wipro's RORE of 15.76%, while respectable, is significantly lower than TCS. This indicates that Wipro may be less efficient in reinvesting its earnings or facing challenges in generating returns from its retained capital.
- **Infosys:** Infosys's RORE of 47.13%, while respectable, is significantly is than Wipro. This indicates that infosys can be efficient in reinvesting its earnings to drive growth and profitability.

Factors Affecting RORE:

- **Profitability:** Higher net income relative to retained earnings leads to a higher RORE.
- **Investment Efficiency:** Effective investment of retained earnings can boost RORE.
- **Dividend Payout:** A higher dividend payout ratio can reduce

18 FACTORS ANALYSIS

FACTORS	TCS	RANKING	SCORE	WIPRO	RANKING	SCORE	INFOSYS	RANKING	SCORE
1.EPS RATIO EPS = (Net Income - Preferred Dividends) / Average Outstanding Share	119.4	1	3	16.63	2	2	0.082	3	1
2.DEBT TO EQUITY RATIO Debt-to-Equity Ratio = Total Liabilities / Total Shareholders' Equity	0.596	2	2	0.319	1	3	26.41	3	1
3.ROA RATIO ROA = Net Income / Average Total Assets	30.03	2	2	11.16	3	1	34.49	1	3
4.CURRENT									

RATIO Current Ratio = Current Assets / Current Liabilities	2.52	3	1	35.78	2	2	36.55	1	3
5.ROCE RATIO ROCE = (EBIT / Average Capital Employed) * 100	30.0 3	1	3	11.16	3	1	19.95	2	2
6.ROE RATIO ROE = (Net Income / Average Shareholders' Equity) * 100	59.2 2	1	3	15.76	3	1	29.47	2	2
7.NET PROFIT MARGIN RATIO Net Profit Margin = (Net Profit / Net Sales) * 100	21.5 3	2	2	5.36	3	1	22.47	1	3
8.PRICE TO EARNING RATIO P/E Ratio = Market Price per Share / Earnings per Share	63.8 6	3	1	15.07	2	2	9.86	1	3
9.DIVIDEND PAYOUT RATIO Dividend Payout Ratio = (Dividends Paid / Net Income) * 100	57.7 0	1	3	5.9	2	2	1.65	3	1
10. PRICE TO BOOK	17.5	2	2	2.71	3	3	27.90	1	3

RATIO P/B Ratio = Market Price per Share / Book Value per Share	3								
11.PRICE TO EARNING GROWTH PEG Ratio = P/E Ratio / Expected Earnings Growth Rate	27.6 5	3	1	1.66	1	3	2.62	2	2
12.GROSS PROFIT RATIO Gross Profit Ratio = (Gross Profit / Net Sales) * 100	25.3 3	3	1	39.25	2	2	44.46	1	3
13.WORKING CAPITAL RATIO Working Capital Ratio = Current Assets / Current Liabilities	2.20	3	1	35.78	1	3	2.63	2	2
14.OPERATING RATIO Operating Ratio = (Operating Expenses / Net Sales) * 100	74.6 3	2	2	82.23	1	3	32.41	3	1
15.QUICK RATIO Quick Ratio = (Cash + Marketable Securities + Accounts Receivable) / Current Liabilities	2.12	3	1	35.78	1	3	9.86	2	2
16.INTEREST									

COVERAGE RATIO ICR = EBIT / Interest Expense	33.84	1	3	11.24	2	2	1.65	3	1
17.ROI RATIO ROI = (Net Profit / Initial Investment) * 100	30.03	1	3	23.57	2	2	11.16	3	1
18.RORE RATIO RORE = (Net Income / Retained Earnings (Beginning)) * 100	60.73	1	3	15.76	3	1	47.13	2	2

Factor Analysis

TCS			WIPRO			INFOSYS		
	RANKING	SCORE		RANKING	SCORE		RANKING	SCORE
119.4	1	3	16.63	2	2	0.082	3	1
0.596	2	2	0.319	1	3	26.41	3	1
30.03	2	2	11.16	3	1	34.49	1	3
2.52	3	1	35.78	2	2	36.55	1	3
30.03	1	3	11.16	3	1	19.95	2	2
59.22	1	3	15.76	3	1	29.47	2	2

21.53	2	2	5.36	3	1	22.47	1	3
63.86	3	1	15.07	2	2	9.86	1	3
57.70	1	3	5.9	2	2	1.65	3	1
17.53	2	2	2.71	3	3	27.90	1	3
27.65	3	1	1.66	1	3	2.62	2	2
25.33	3	1	39.25	2	2	44.46	1	3
2.20	3	1	35.78	1	3	2.63	2	2
74.63	2	2	82.23	1	3	32.41	3	1
2.12	3	1	35.78	1	3	9.86	2	2
33.84	1	3	11.24	2	2	1.65	3	1
30.03	1	3	23.57	2	2	11.16	3	1
60.73	1	3	15.76	3	1	47.13	2	2
TOTAL		40	TOTAL		37	TOTAL		32

Interpretation:

Overall Ranking

Based on the total scores across all three categories, TCS seems to be the top-performing company, followed by Wipro and then Infosys.

Category-wise Performance:

Ranking 1

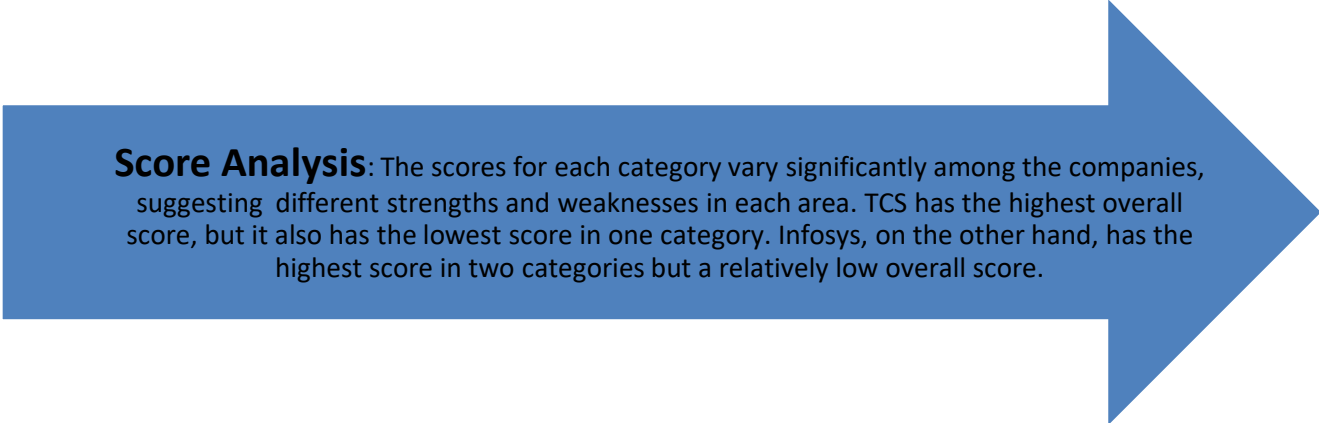
TCS consistently holds the top position in this category, indicating strong performance in the first criterion. Wipro and Infosys occupy the second and third positions, respectively.

Ranking 2

Infosys excels in this category, securing the first rank. Wipro and TCS follow in the second and third positions

Ranking 3

Infosys maintains its dominance in this category, while TCS and Wipro share the second and third positions.



Score Analysis: The scores for each category vary significantly among the companies, suggesting different strengths and weaknesses in each area. TCS has the highest overall score, but it also has the lowest score in one category. Infosys, on the other hand, has the highest score in two categories but a relatively low overall score.

Additional Considerations:

Weighting: The data doesn't specify the weight assigned to each category. If the categories have different importance levels, the overall ranking might change.

Data Source and Methodology: The reliability of the rankings depends on the source of the data and the methodology used to calculate the scores. Without this information, it's difficult to assess the accuracy of the results.

Conclusion:

While TCS appears to be the overall leader based on the given data, a more comprehensive analysis considering factors like weighting and data source is necessary to draw definitive conclusions about the performance of these companies.

CHAPTER 6

CONCLUSION

Financial Performance:

- **Profitability:** Both TCS and Wipro have demonstrated strong profitability, as evidenced by their healthy ROCE, ROE, and net profit margins. However, TCS has consistently outperformed Wipro in terms of profitability.
- **Efficiency:** TCS has shown superior efficiency in managing its assets and capital, as reflected in its higher ROCE and ROA.
- **Financial Health:** Both companies have strong financial positions, with solid liquidity ratios and interest coverage ratios. However, TCS's higher interest coverage ratio indicates a stronger financial cushion.

Valuation:

- **P/E Ratio:** Wipro's P/E ratio is significantly higher than TCS's, suggesting that investors are willing to pay a premium for Wipro's future growth.
- **P/B Ratio:** TCS has a higher P/B ratio, indicating that investors may perceive its assets as more valuable.
- **Dividend Payout:** TCS has a higher dividend payout ratio, suggesting a greater focus on returning value to shareholders.

Overall Assessment:

- **TCS:** TCS has consistently outperformed Wipro in terms of profitability and efficiency. Its higher valuation might be justified by its strong financial performance and growth prospects.
- **Wipro:** Wipro has demonstrated solid financial performance, but its valuation is relatively lower than TCS. This may present an opportunity for investors seeking a potentially undervalued stock.

Important Considerations:

- **Industry Dynamics:** The IT services industry is highly competitive, and both TCS and Wipro face similar challenges and opportunities.
- **Economic Factors:** Economic conditions and industry trends can significantly impact the performance of both companies.
- **Individual Investor Preferences:** Investors should consider their own risk tolerance, investment objectives, and time horizon when evaluating TCS and Wipro.

In conclusion, both TCS and Wipro are strong companies in the IT services industry. However, TCS appears to have a stronger overall financial profile and valuation, while Wipro may offer a potentially more attractive valuation.

It's essential to conduct thorough research and consider multiple factors before making investment decisions.

CHAPTER 7 - LIMITATIONS OF THE STUDY

- **1. Subjectivity of Analysis:**

While the analysis is based on objective data, the interpretation and conclusions drawn can vary depending on the analyst's perspective and expertise. Different analysts may have different insights and recommendations.

- **2. Time Factor:**

The analysis does not consider the time required to complete specific tasks or projects. This can impact the overall efficiency and profitability of the companies.

- **3. Lack of Direct Stakeholder Interaction:**

The analysis is based on publicly available information and may not capture the nuances of the companies' operations or the perspectives of key stakeholders. Direct interaction with stakeholders could provide valuable insights that are not reflected in the data.

- **4. Potential for Errors and Oversights:**

There is always a risk of errors or misinterpretations in any analysis, regardless of the analyst's expertise. Careful review and cross-verification can help minimize these risks.

CHAPTER 8

BIBLIOGRAPHY

- <https://in.linkedin.com/company/tata-consultancy-services>
- Fundamental analysis and position trading : evaluation of trader
- Reference books/ journals
- <https://www.investopidea.com/>
- <https://www.moneycontrol.com/india/>
- www.screener.com
- [LinkedIn](#)
- www.tcs
- https://en.wikipedia.org/wiki/Tata_Consultancy_Services
- www.infosys
- <https://www.infosys.com/>
- www.wipro
- <https://en.wikipedia.org/wiki/Wipro>